

Statistical Modelling

Derek Aguiar

University of Connecticut – CSE5825

Announcements

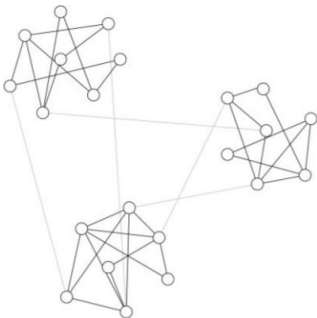
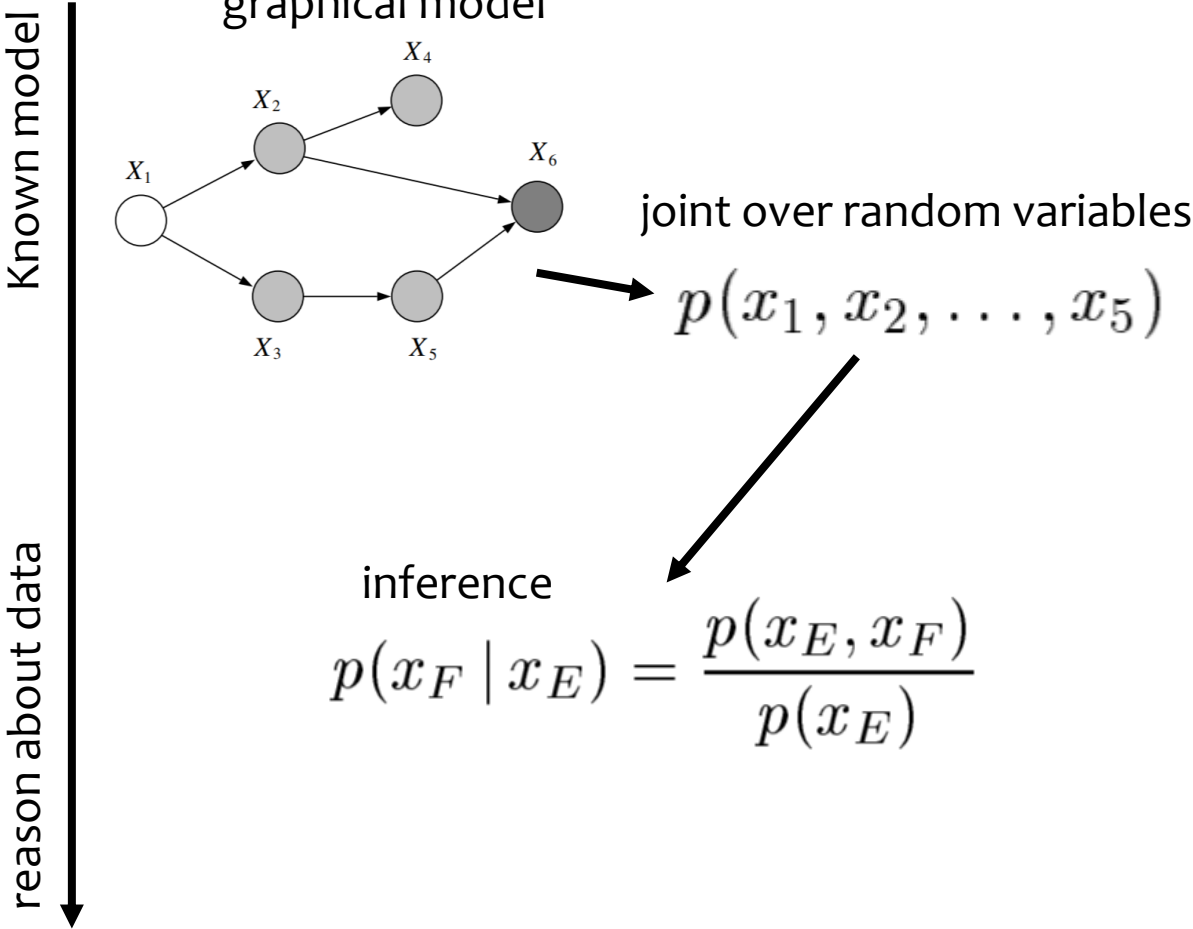
-

Statistical Concepts

• Probabilist

vs.

statistician

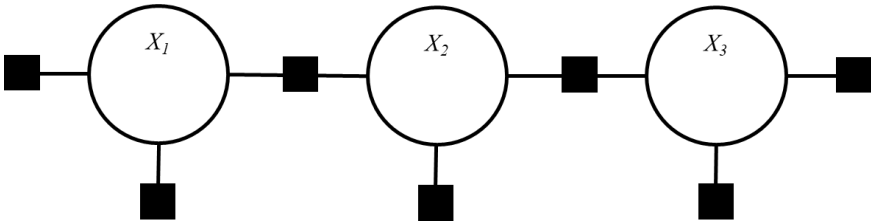


Known data

```
CTCTTGGCTCCAGCATCGATGAAGAACGCZ
ATTTAGAGGAAGTAAAAGTCGTAAACAAGGT
ACTGTCAAAACTTTTAACAACGGATCTCTT
TTGCTTCGGCGGGCGCCCGCAAGGGTGCCCC
CCTGCCGTGGCAGATCCCCAACGCCGGGCC
TCTTGGCTCCAGCATCGATGAAGAACGCAC
GCATCGATGAAGAACGCAGCGAAACGCCAT
ATACTTCTGAGTGTTCTTAGCGAACTGTCZ
GATCTCTTGGCTCCAGCATCGATGAAGAAC
```

What are the communities?

Is this a gene?



reason about model

Central question for statistical modelling

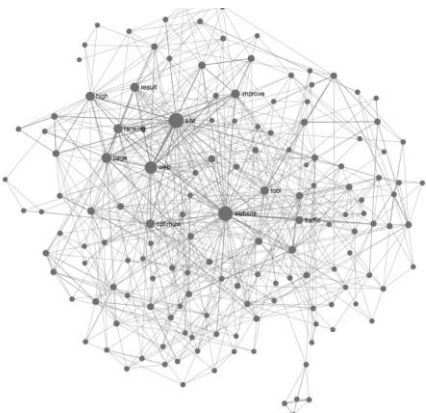
- **How can we use joint probability distributions to make meaningful statements about observed data?**

Probabilistic modelling

	Statistical interpretation	Graphical interpretation
Data	observed random variables, x	shaded nodes
Joint distribution	involves hidden random variables $p(x, z)$	inference on unshaded nodes $p(x, z)$
Conditional dist.	informs about the data, enables predictions	informs about the data, enables predictions

Examples

Observed data



Model

$$p((x,y) \in E) \propto \exp \left\{ \underbrace{\sum_{C_k \supseteq \{x,y\}} \langle \phi(x,y), \theta_k \rangle}_{\text{circles containing both nodes}} - \underbrace{\sum_{C_k \not\supseteq \{x,y\}} \alpha_k \langle \phi(x,y), \theta_k \rangle}_{\text{all other circles}} \right\}.$$

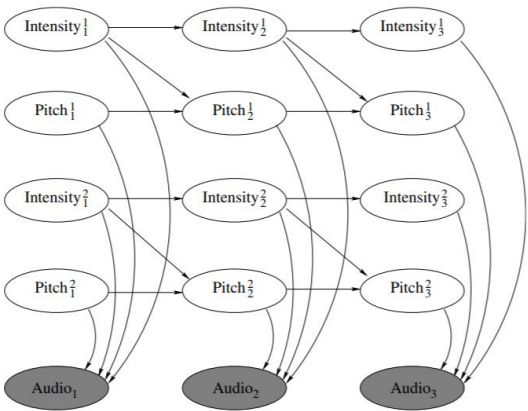
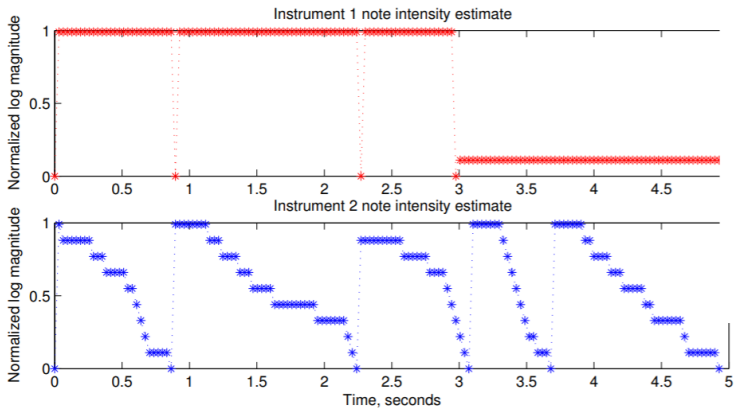
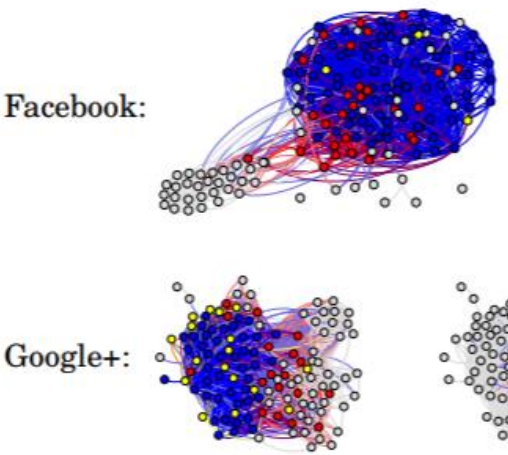


Fig. 1. A DGM for two-instrument transcription. Shaded nodes are observed data; unshaded nodes are “hidden” or “latent” variables.

Hidden random variables



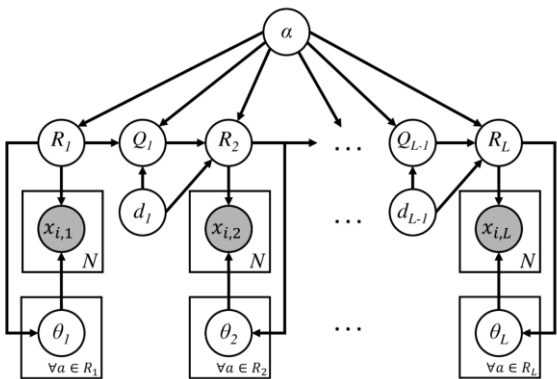
Examples

Observed data

CTCTTGGCTCCAGCATCGATGAAGAACGC
ATTTAGAGGAAGTAAAAGTCGTAACAAGGT
ACTGTCAAAACTTTTAACAACGGATCTCTT
TTGCTTCGGCGGGCGCCCGCAAGGGTGCCCC
CCTGCCGTGGCAGATCCCCAACGCCGGGCC
TCTTGGCTCCAGCATCGATGAAGAACGCAC
GCATCGATGAAGAACGCAGCGAAACGCGAT
ATACTTCTGAGTGTTCTTAGCGAACTGTCA
GATCTCTTGGCTCCAGCATCGATGAAGAAC

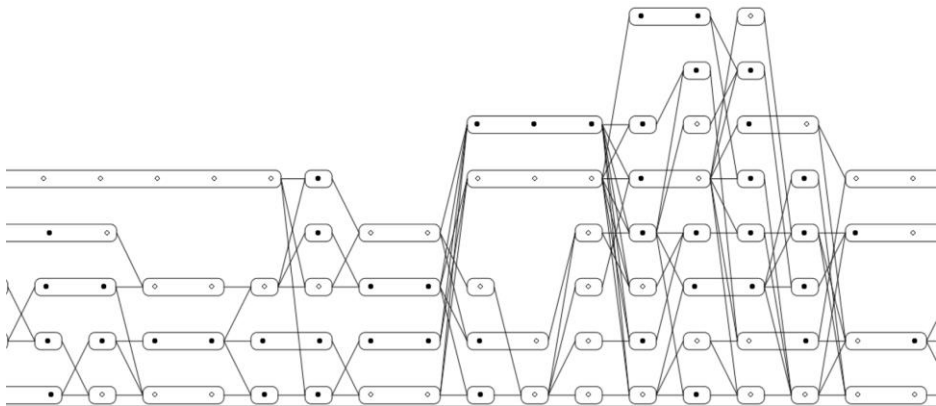
?

Model



?

Hidden random variables



?

Prob and Stats

- Discrete random variables



- Continuous random variables

- Density $p(x)$, if $x \in \mathbb{R}$, then $\int_{-\infty}^{\infty} p(x)dx = 1$
- Probabilities are integrals over intervals
 - $P(X \in (-1,1)) = \int_{-1}^1 p(x)dx$



Example: The Gaussian distribution

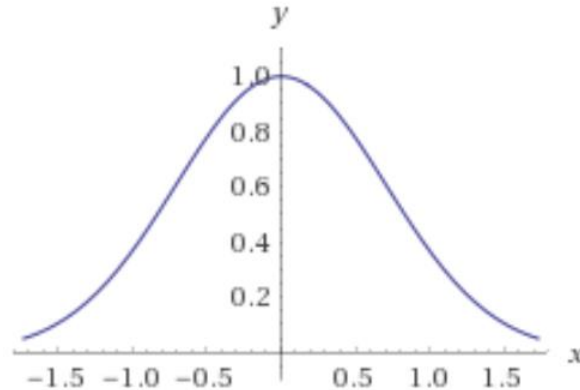
- The Gaussian (Normal) distribution density is

- The density is proportional $p(x | \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}$
- Assume $\mu = 0, \sigma = 1$

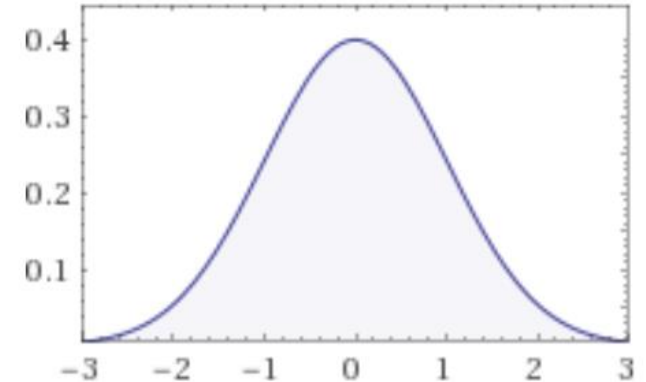
Parameters

$$\begin{array}{l} \mu \\ \sigma^2 > 0 \end{array}$$

unnormalized



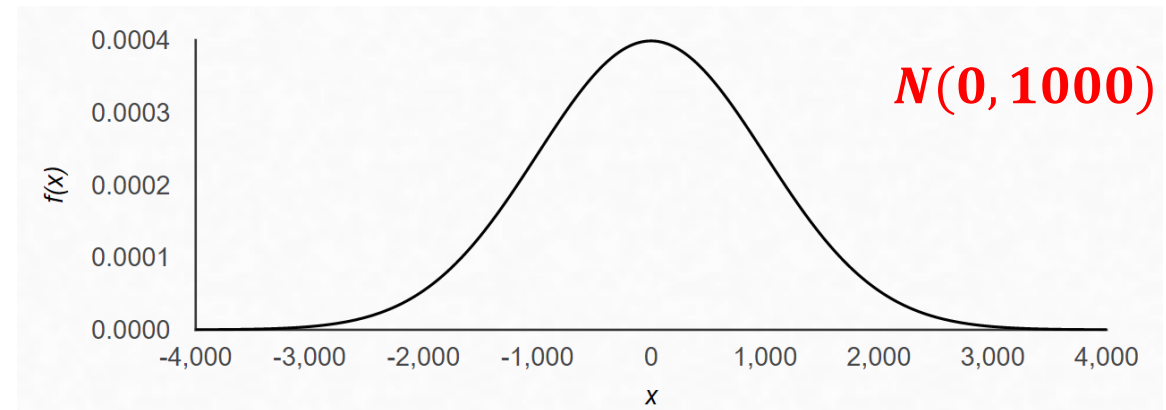
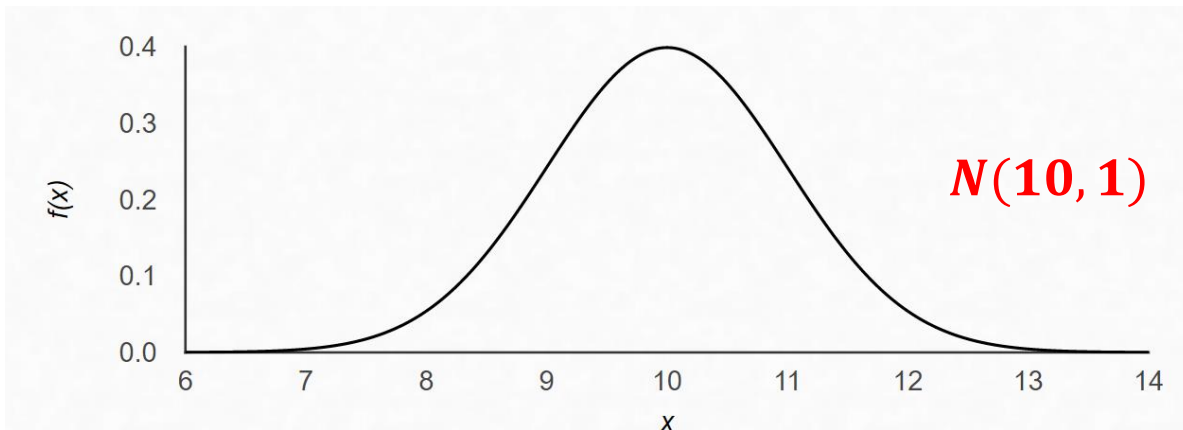
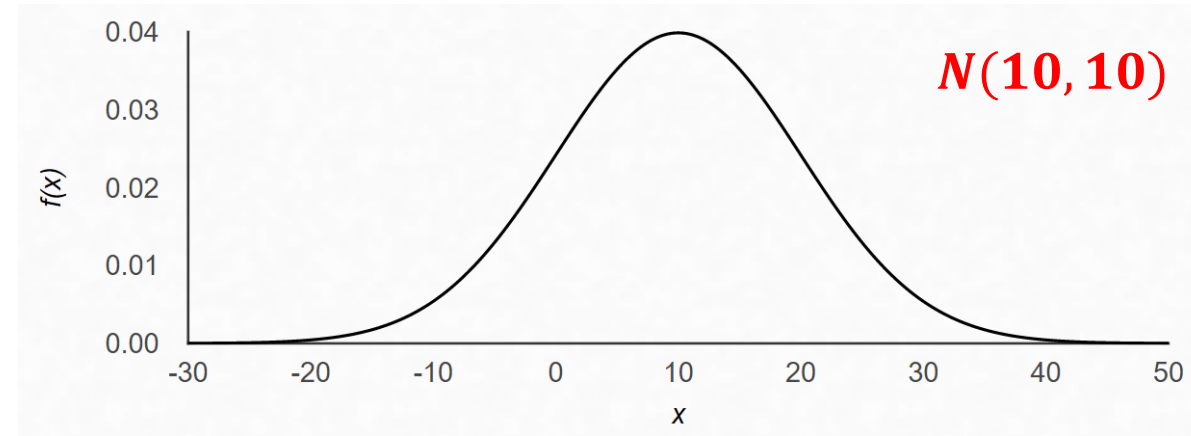
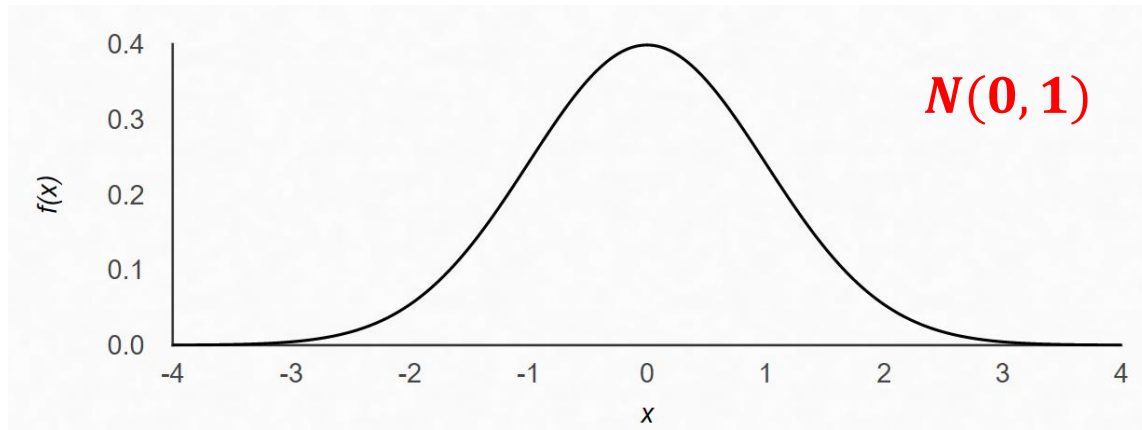
$N(\mu = 0, \sigma^2 = 1)$



Parameters

e.g. Bernoulli random variable

- Parameters are values that index a family of distributions $p(x|\theta) = \theta^{1[x=H]}(1 - \theta)^{1[x=T]}$
- Gaussian has two parameters, mean and variance



Parameters (Beta)

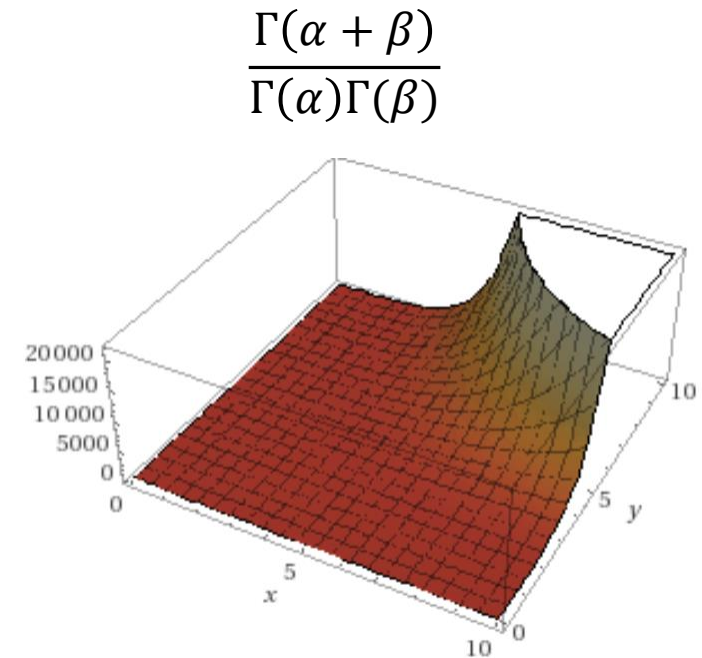
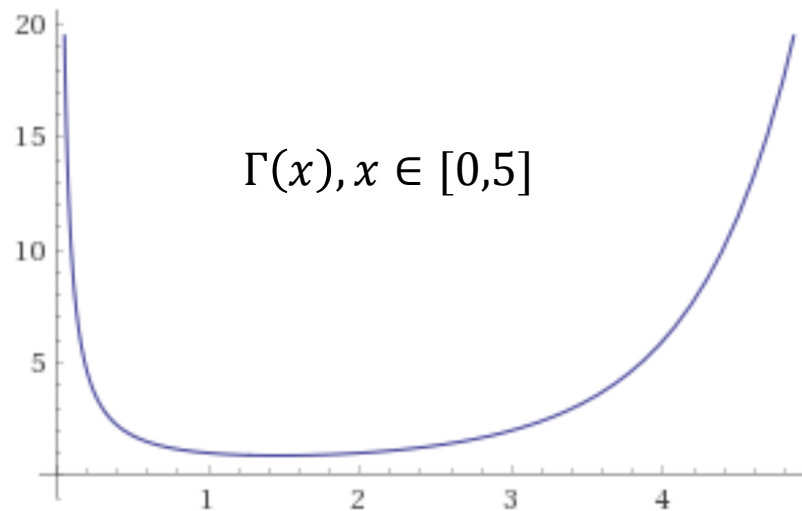
- A beta random variable is over $(0,1)$ with density

$$p(\pi | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

- Parameters are $\alpha > 0$ and $\beta > 0$

- Where $\Gamma(x) = (x - 1)!$
if x is integer

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx$$



Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (x-1)!$$

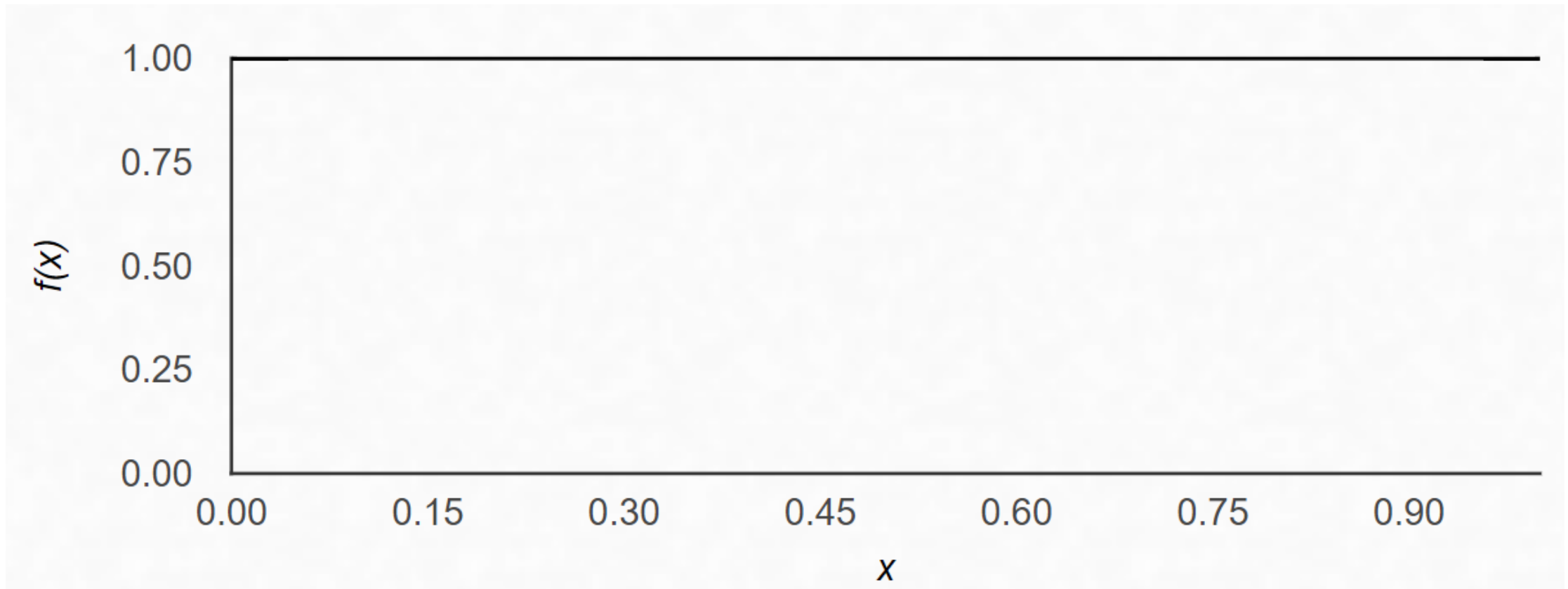
$$\textit{Beta}(\alpha = 1, \beta = 1)$$

Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n - 1)!$$

$$\textit{Beta}(\alpha = 1, \beta = 1)$$



Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n-1)!$$

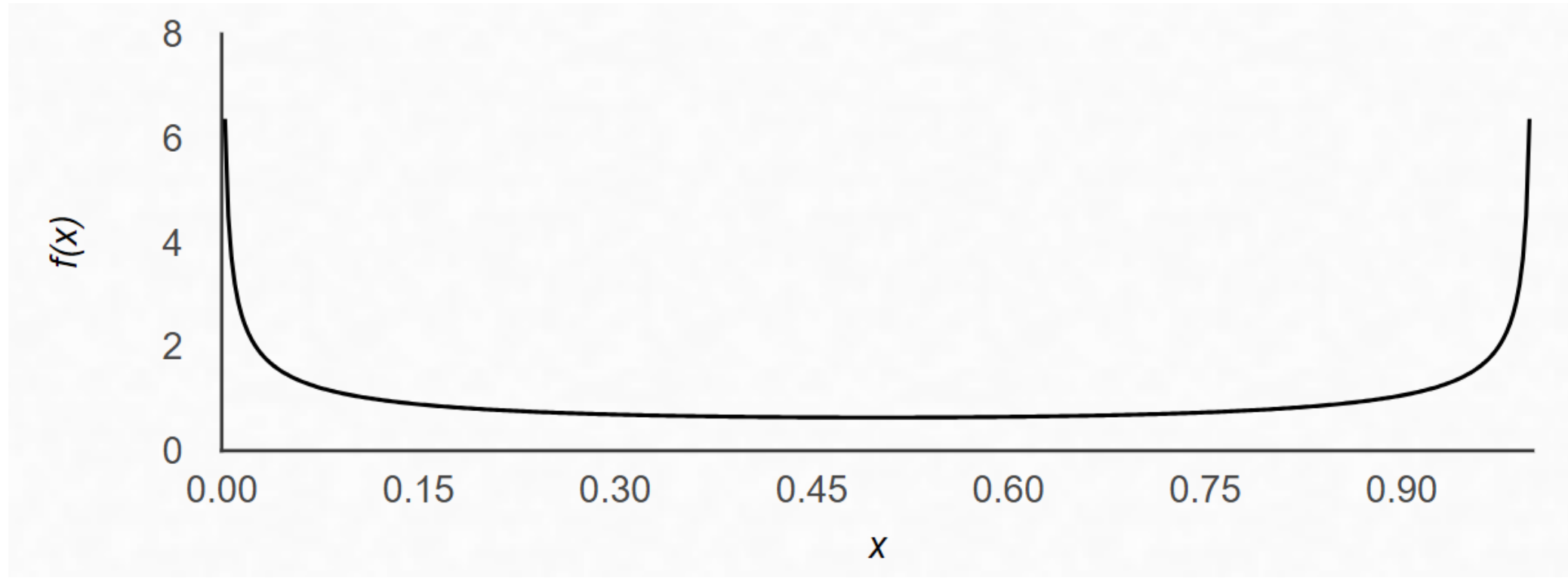
$$\textit{Beta}(\alpha = 0.5, \beta = 0.5)$$

Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n - 1)!$$

$$\text{Beta}(\alpha = 0.5, \beta = 0.5)$$



Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n-1)!$$

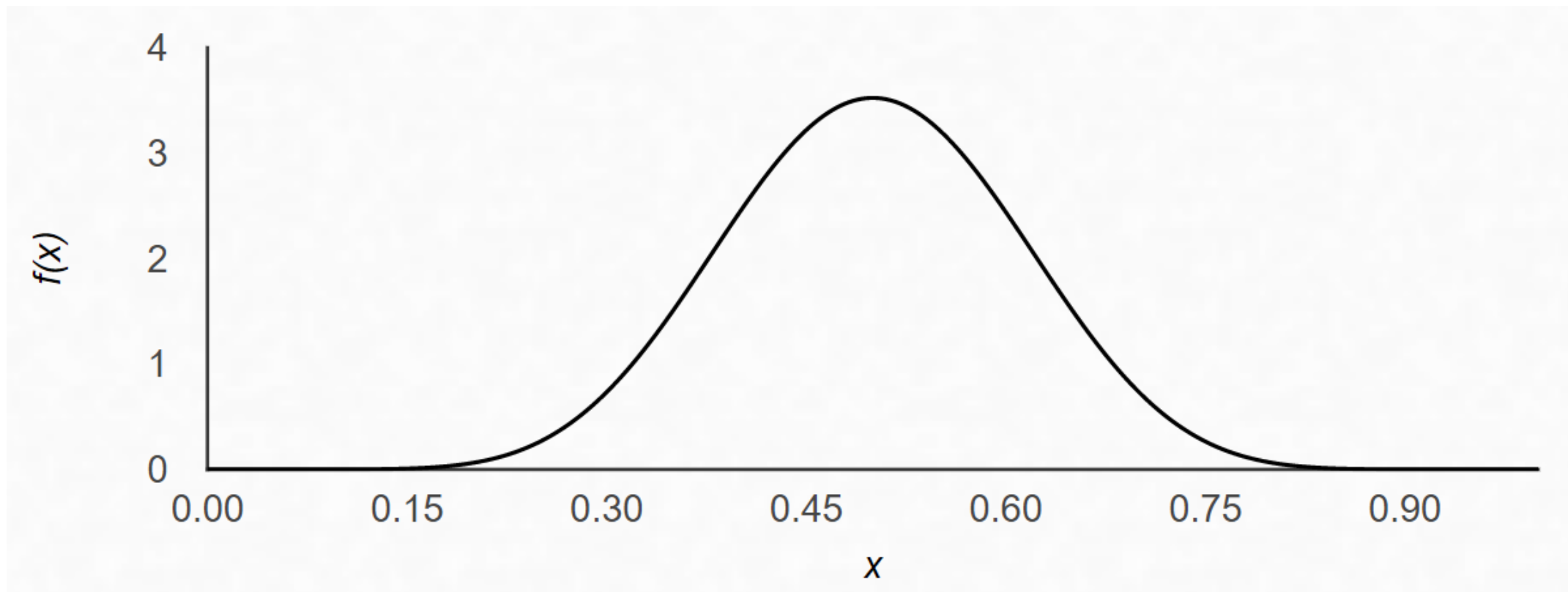
$$\textit{Beta}(\alpha = 10, \beta = 10)$$

Beta distribution

$$p(\pi | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n-1)!$$

$$\text{Beta}(\alpha = 10, \beta = 10)$$



Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n-1)!$$

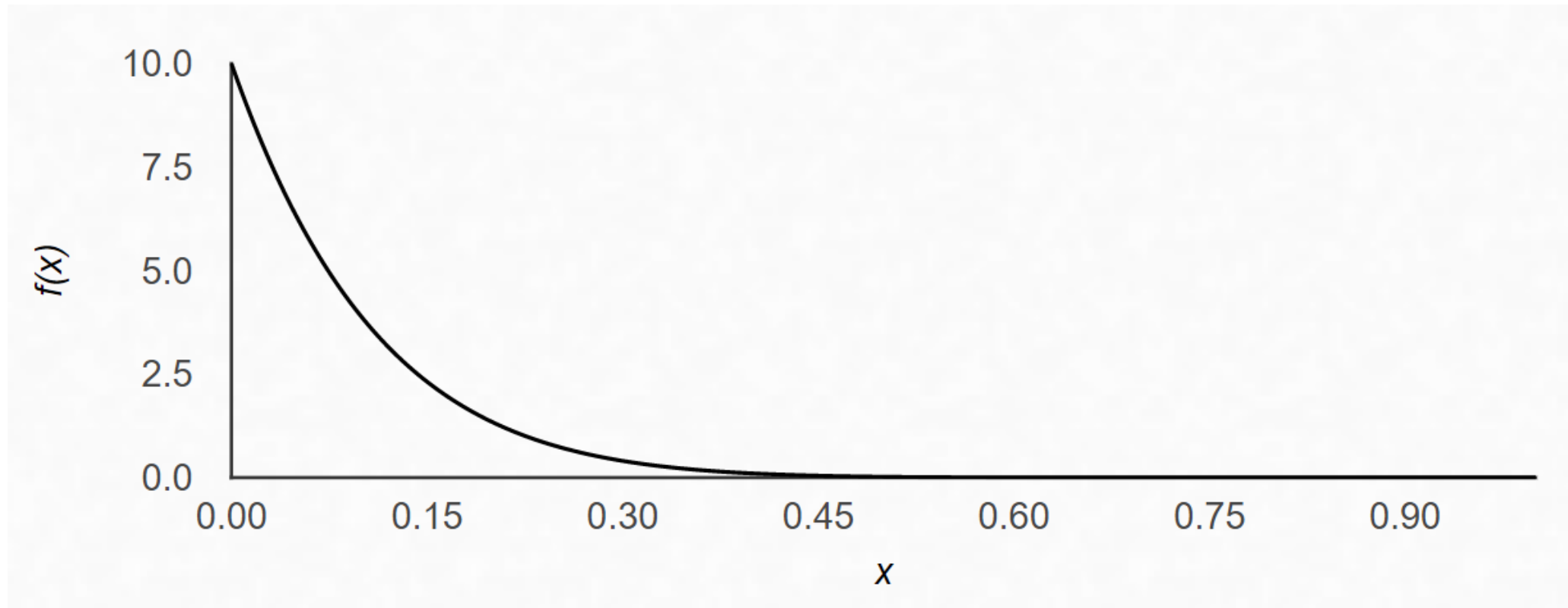
$$\textit{Beta}(\alpha = 1, \beta = 10)$$

Beta distribution

$$p(\pi | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n - 1)!$$

$$\text{Beta}(\alpha = 1, \beta = 10)$$



Beta distribution

$$p(\pi \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n-1)!$$

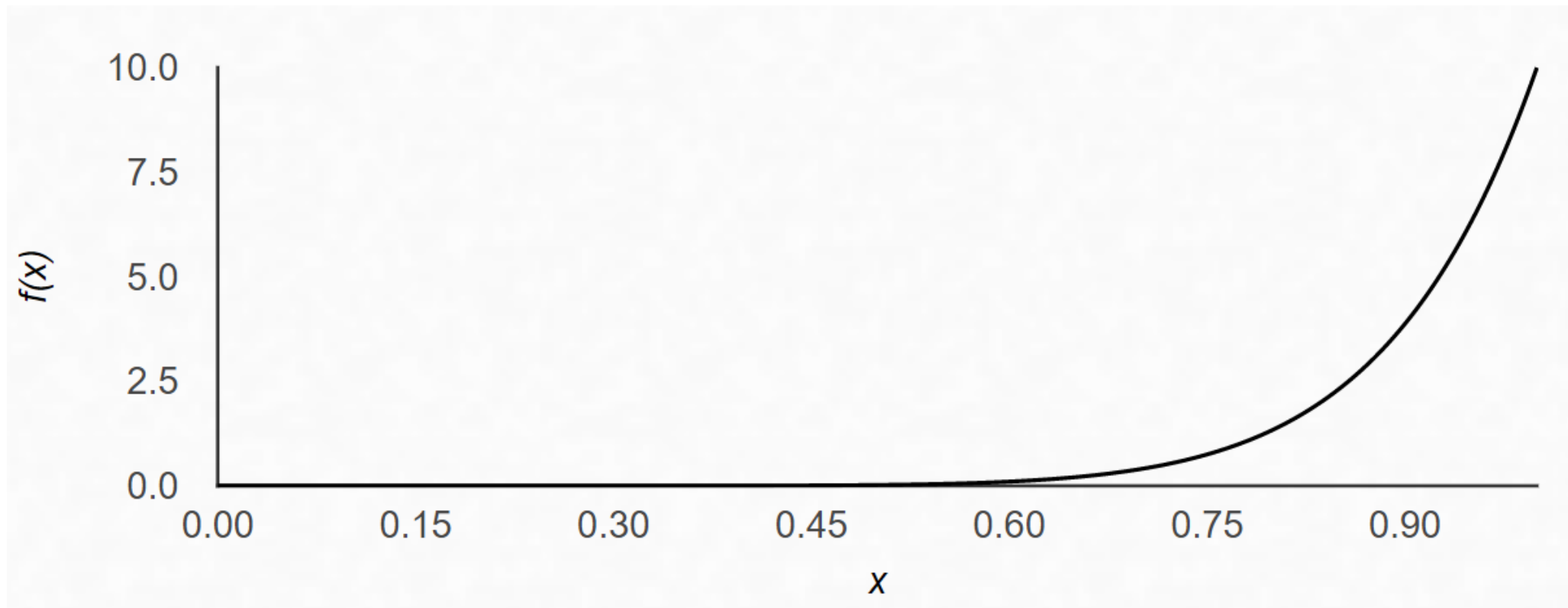
$$Beta(\alpha = 10, \beta = 1)$$

Beta distribution

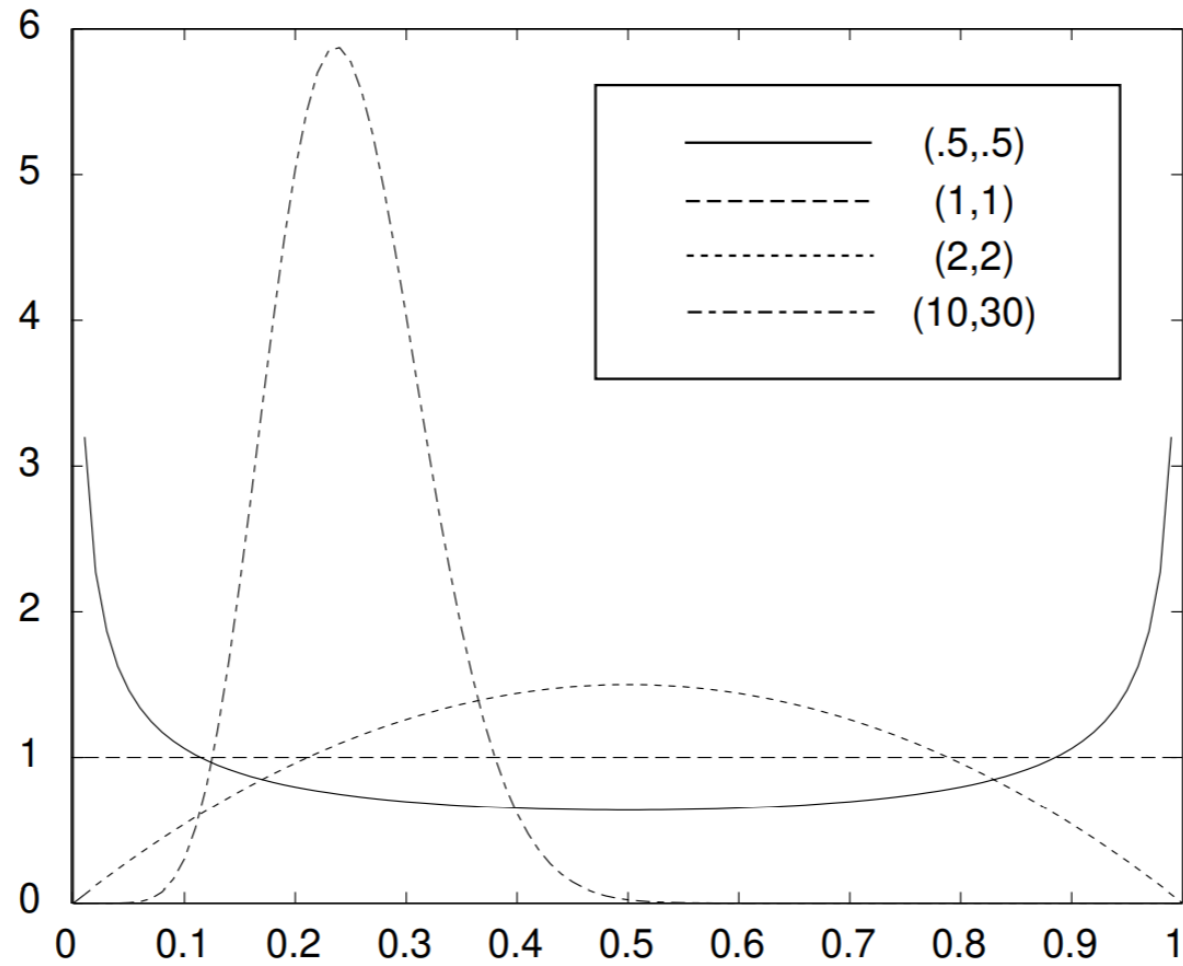
$$p(\pi | \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1}$$

$$\Gamma(x) = (n - 1)!$$

$$\text{Beta}(\alpha = 10, \beta = 1)$$



Beta distribution



Relationship between beta and Bernoulli

- A draw from a Bernoulli is what?

Relationship between beta and Bernoulli

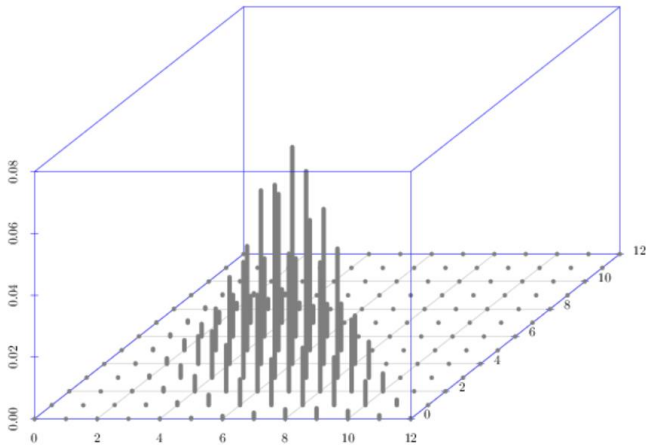
- A draw from a Bernoulli is what?
- What determines the probability of H/T?

Relationship between beta and Bernoulli

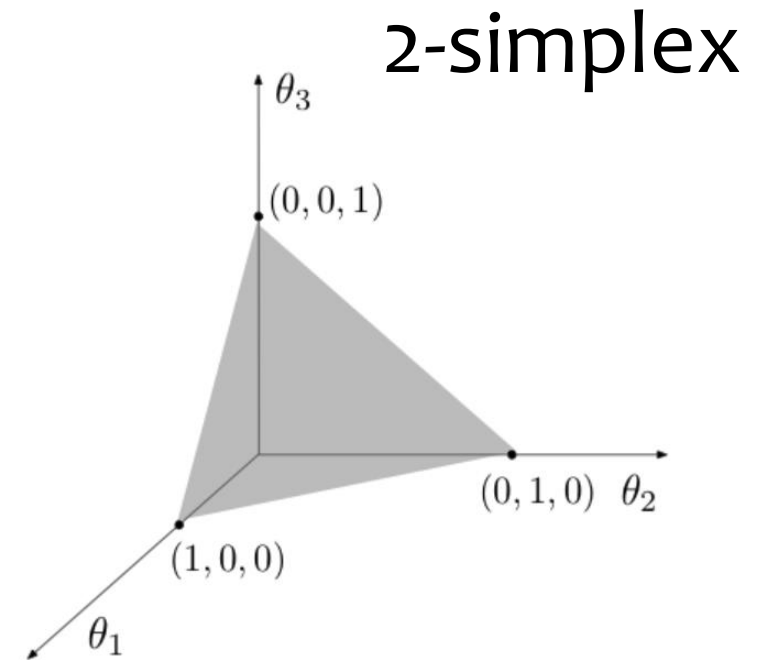
- A draw from a Bernoulli is what?
- What determines the probability of H/T?
- A draw from a beta is what?

Multinomial distribution

- A multinomial (or categorical) distribution has a vector parameter on the $(k - 1)$ -simplex.
- The $(k - 1)$ -simplex is the space of non-negative vectors of length k that sum to one.
 - $\theta_i > 0$ and $\sum_i \theta_i = 1$



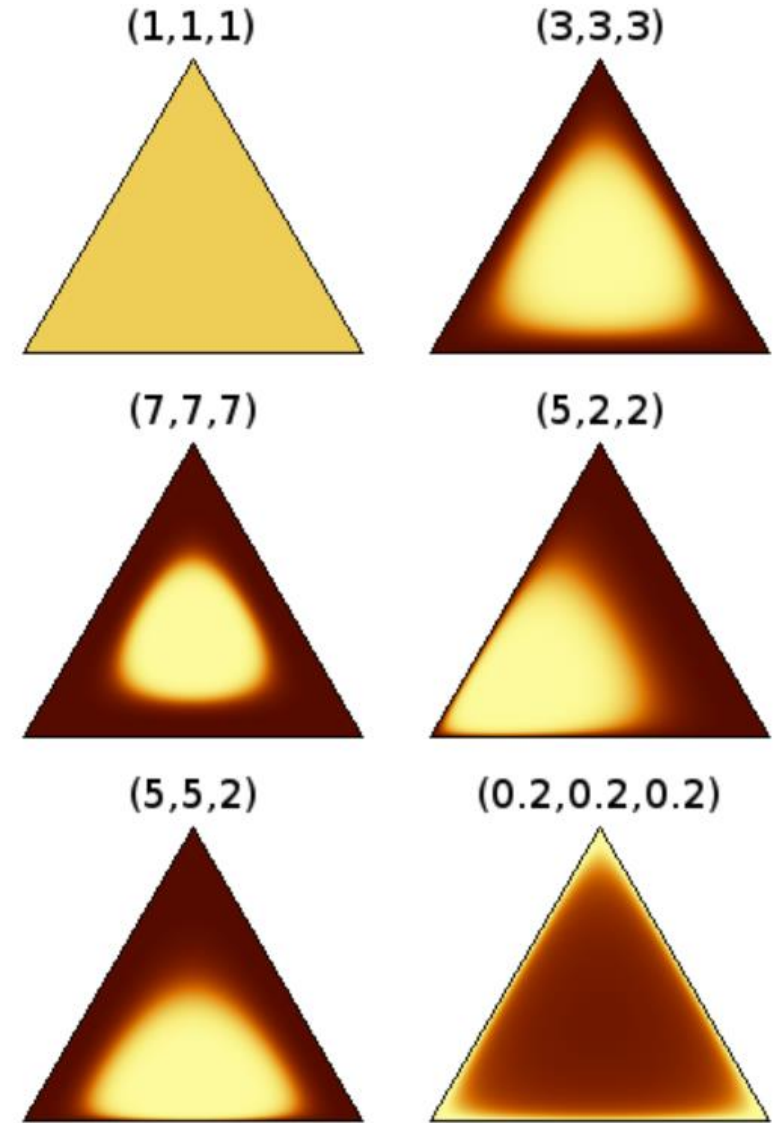
$$\text{Mult}(n, \boldsymbol{\theta}) = \frac{n!}{x_1! \dots x_k!} \theta_1^{x_1} \dots \theta_k^{x_k}$$



Dirichlet distribution

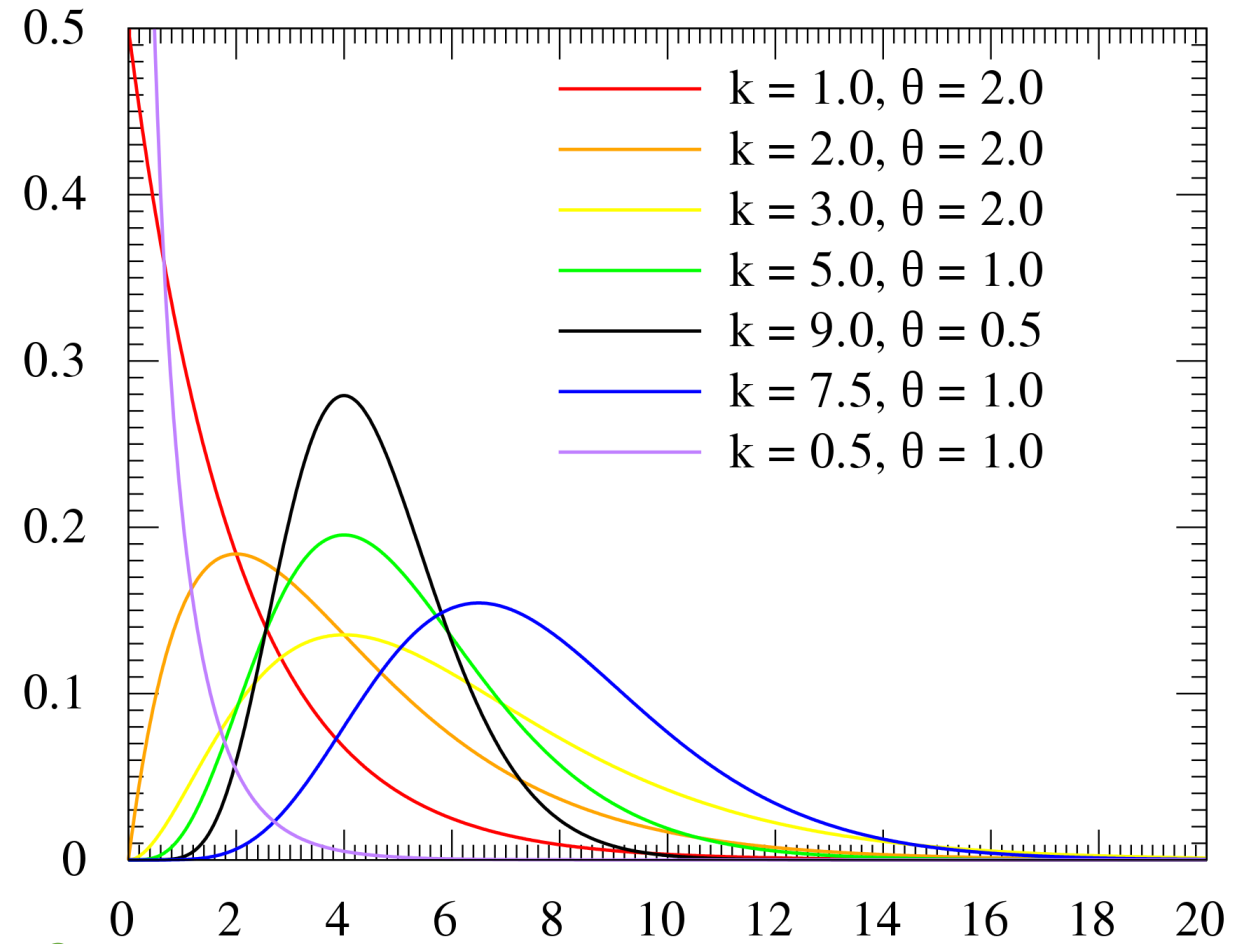
- The categorical distribution is the vector generalization of the Bernoulli
- The vector generalization of the beta distribution is the *Dirichlet* distribution
- It is a distribution on the $(k - 1)$ -simplex parameterized by a k -vector $\alpha_j > 0$ with density

$$\frac{\Gamma\left(\sum_{j=1}^k \alpha_j\right)}{\prod_{j=1}^k \Gamma(\alpha_j)} \prod_{j=1}^k \theta_j^{(\alpha_j-1)}$$



Gamma Distribution

$$\frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-\frac{x}{\theta}}$$



What kind of random variables can we model with gamma?

Expectation

- Let X be a random variable.
 - $f(X)$ is also a random variable.
 - The expectation is a weighted average of f determined by $p(x)$

$$\mathbb{E}[f(X)] = \sum_x p(x) f(x).$$

$$\mathbb{E}[f(X)] = \int p(x) f(x) dx.$$

- The conditional expectation is defined similarly

$$\mathbb{E}[f(X) | Y = y] = \sum_x p(x | y) f(x).$$

What is this?
A number?
A distribution?

Expectation

- Expectation is a function of parameters
 - If $X \sim N(\mu, \sigma^2)$ then $E[X] = \mu$
 - If $X \sim \text{Beta}(\alpha, \beta)$ then $E[X] = \frac{\alpha}{\alpha + \beta}$
- Expectations are often useful summaries for distributions
 - $X = \text{height}, Y = \text{weight}, E[X|Y]$ or $E[Y|X]$
- The second moment, variance, is also useful for characterizing a distribution
$$\text{Var}[X] = E[X^2] - E[X]^2$$

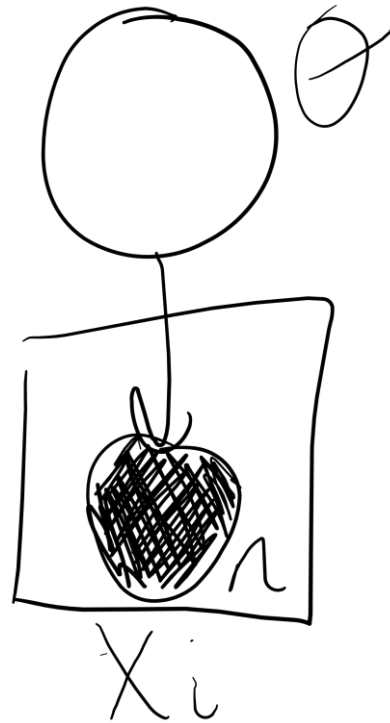
Beta distribution variance is $\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$

Statistical Problems

- In a statistical model, data are observed random variables which are dependent on hidden random variables
- Example, n observations from a Bernoulli with unknown θ

Statistical Problems

- In a statistical model, data are observed random variables which are dependent on hidden random variables
- Example, n observations from a Bernoulli with unknown θ



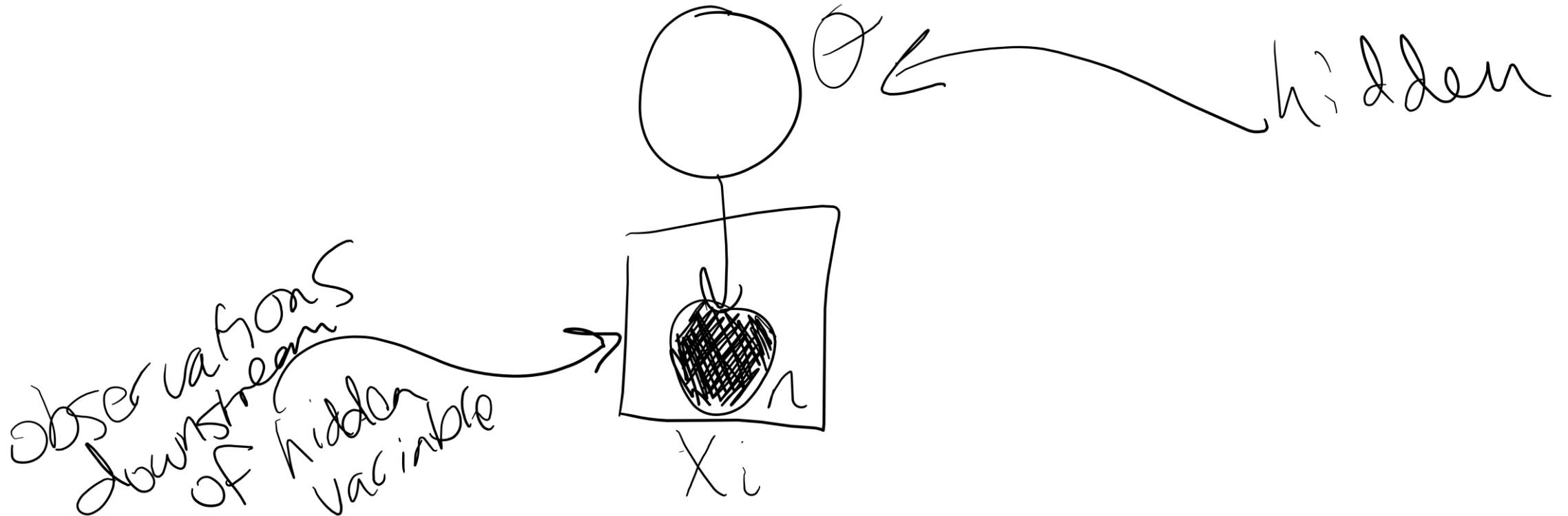
Statistical Problems

- In a statistical model, data are observed random variables which are dependent on hidden random variables
- Example, n observations from a Bernoulli with unknown θ



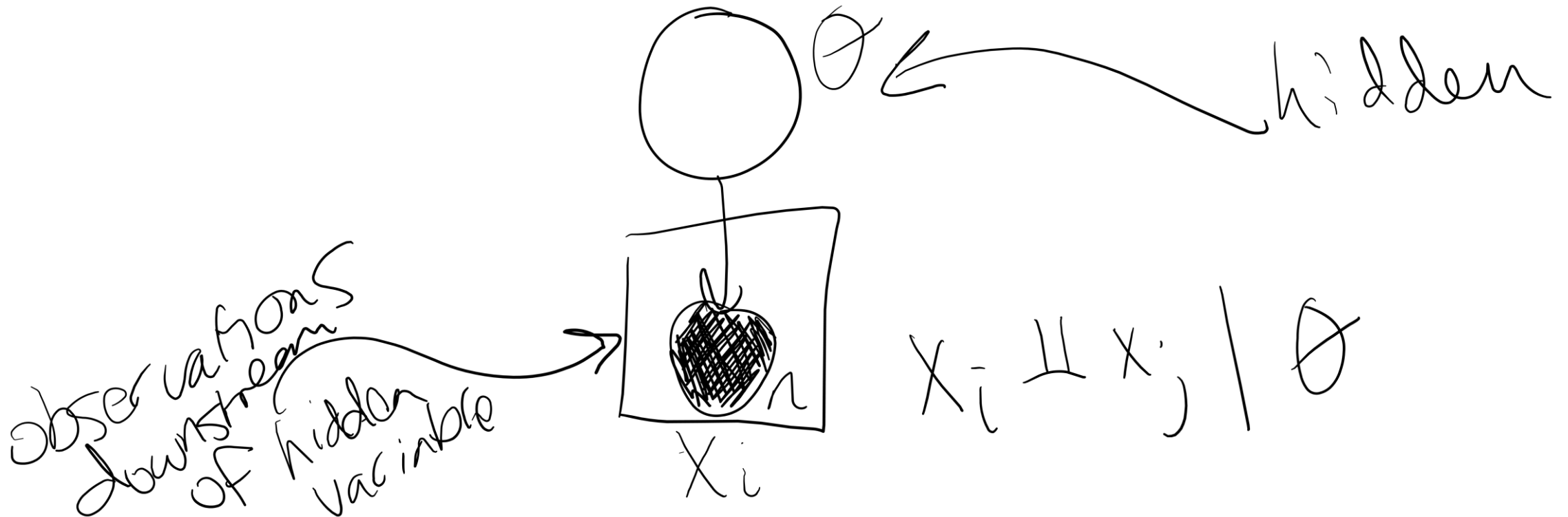
Statistical Problems

- In a statistical model, data are observed random variables which are dependent on hidden random variables
- Example, n observations from a Bernoulli with unknown θ



Statistical Problems

- In a statistical model, data are observed random variables which are dependent on hidden random variables
- Example, n observations from a Bernoulli with unknown θ



Specifying a statistical model

1. Graphical model

$$p(\pi, x_{1:n}) = p(\pi) \prod_{i=1}^n p(x_i \mid \pi).$$

- Conditional independence

distributed?

2. Joint distribution

- Full specified conditional independencies and joint factorization
- We know that x 's are Bernoulli r.v.s, what about π ?

Specifying a statistical model

1. Graphical model

$$p(\pi, x_{1:n}) = p(\pi) \prod_{i=1}^n p(x_i \mid \pi).$$

- Conditional independence

distributed?

2. Joint distribution

- Full specified conditional independencies and joint factorization

$$p(\pi, x_{1:n}) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1} \prod_{i=1}^n \pi^{x_i} (1 - \pi)^{1-x_i}$$

Generative model for discrete model

3) Probabilistic generative process

- Intuitive, relates to data generation
- Choose $\pi \sim \text{Beta}(\alpha, \beta)$
- For each data point i :
 - Choose $x_i \mid \pi \sim \text{Bern}(\pi)$
- It's easy to write a simulator for this process (and quite useful for research purposes) and inference amounts to reversing this generative process to find $p(\pi \mid x)$

The posterior

- We are interested in $p(\theta|x)$: what are the values of the quantities used to generate my data?
- We have a statistical model of how data is generated
- Let's work through our simple model

Posterior of simple beta-Bernoulli model

- Let $x = 0$ denote tails, $x = 1$ denote heads

$$p(\pi \mid x_{1:n})$$

Posterior of simple beta-Bernoulli model

- Let $x = 0$ denote tails, $x = 1$ denote heads

$$\begin{aligned} p(\pi \mid x_{1:n}) &\propto p(\pi) \prod_{i=1}^n p(x_i \mid \pi) \\ &= \left[\frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1} \right] \left[\prod_{i=1}^n \pi^{x_i} (1 - \pi)^{1-x_i} \right] \\ &\propto \left[\pi^{\alpha-1} (1 - \pi)^{\beta-1} \right] \left[\pi^{(\sum_{i=1}^n x_i)} (1 - \pi)^{(\sum_{i=1}^n (1-x_i))} \right] \\ &= \pi^{(\alpha-1+\sum_{i=1}^n x_i)} (1 - \pi)^{(\beta-1+\sum_{i=1}^n (1-x_i))} \end{aligned}$$

Posterior of simple beta-Bernoulli model

- Unscaled beta distribution

$$\pi^{(\alpha-1+\sum_{i=1}^n x_i)} (1-\pi)^{(\beta-1+\sum_{i=1}^n (1-x_i))} = \pi^{\hat{\alpha}-1} (1-\pi)^{\hat{\beta}-1}$$

where

$$\hat{\alpha} = \alpha + \sum_{i=1}^n x_i$$
$$\hat{\beta} = \beta + \sum_{i=1}^n (1 - x_i)$$

The posterior is $Beta(\hat{\alpha}, \hat{\beta})$ [[DEMO](#)]

Some points of discussion

- Maximum likelihood estimation
- Conjugacy
- Posterior expectation
- Posterior predictive distribution
- Penalized likelihood

Maximum likelihood estimation (MLE)

- Classical way of estimating unknown parameters
- MLE is the value of π that maximizes the probability of the data
- Assume: data are *independent and identically distributed* from a Bernoulli with unknown π

$$\hat{\pi}^{MLE} = \operatorname{argmax}_{\pi} p(\mathbf{x}|\pi)$$

Maximum likelihood estimation (MLE)

$$\begin{aligned}\hat{\pi}^{\text{MLE}} &= \arg \max_{\pi} \log p(x_{1:n} \mid \pi) \\ &= \arg \max_{\pi} \sum_{i=1}^n \log p(x_i \mid \pi) \\ &= \arg \max_{\pi} \sum_{i=1}^n (x_i \log \pi_i + (1 - x_i) \log(1 - \pi_i))\end{aligned}$$

Maximum likelihood estimation (MLE)

$$= \operatorname{argmax}_{\pi} \sum_{i=1}^n (x_i \log \pi_i + (1 - x_i) \log(1 - \pi_i))$$

$$\frac{d}{d\pi} \left[\sum_{i=1}^n x_i \log \pi_i + (1 - x_i) \log(1 - \pi_i) \right] = 0$$

Solving for π gives us $\hat{\pi}_{MLE} = \frac{n_{heads}}{n}$

Conjugacy

- In our example,

$$p(\pi) \sim \text{Beta}(\alpha, \beta)$$

$$p(\pi|x) \sim \text{Beta}(\hat{\alpha}, \hat{\beta})$$

- The beta and Bernoulli distributions are conjugate pairs
- Most of our models will not be solved analytically, but conjugacy will be important

Posterior expectation

- We can calculate the expectation of π , known as posterior expectation or *Bayes estimate*

$$\mathbb{E} [\pi \mid x_{1:n}] = \frac{\alpha + \sum_{i=1}^n x_i}{\alpha + \beta + n}$$

- What happens as $n \rightarrow \infty$?

More samples, stronger influence on posterior

- Fewer data $\rightarrow$ prior more important
- More data $\rightarrow$ prior less important

$$\begin{aligned} & \text{Var}[\pi] \\ &= \frac{(\alpha + \sum_{i=1}^n x_i)(\beta + \sum_{i=1}^n (1 - x_i))}{\left((\alpha + \sum_{i=1}^n x_i) + (\beta + \sum_{i=1}^n (1 - x_i))\right)^2 ((\alpha + \sum_{i=1}^n x_i) + (\beta + \sum_{i=1}^n (1 - x_i)) + 1)} \end{aligned}$$

- Grows as $\frac{o(n^2)}{o(n^3)}$
- This does not mean more confidence about the estimate! The posterior variance is a property of the model.

Posterior predictive distribution

- Suppose we want $p(x_{n+1}|x_{1:n})$?
 - How do we compute this?

Posterior predictive distribution

- Suppose we want $p(x_{n+1} | x_{1:n})$?
 - How do we compute this?

$$\begin{aligned} p(x_{n+1} | x_{1:n}) &= \int p(x_{n+1}, \pi | x_{1:n}) d\pi \\ &= \int p(\pi | x_{1:n}) p(x_{n+1} | \pi) d\pi \\ &= \mathbb{E} [p(x_{n+1} | \pi) | x_{1:n}]. \end{aligned}$$

$$\mathbb{E} [f(X) | Y = y] = \sum_x p(x | y) f(x).$$

Penalized likelihood

- Relationship between MLE and MAP
- MAP = value of maximum probability (mode) of posterior

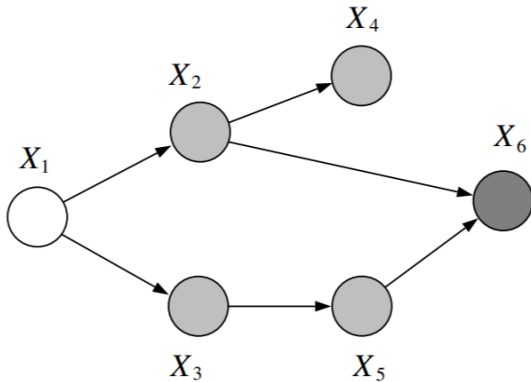
$$\begin{aligned}\hat{\pi}^{\text{MAP}} &= \arg \max_{\pi} \log p(\pi \mid x_{1:n}) \\ &= \arg \max_{\pi} \log p(\pi, x_{1:n}) \\ &= \arg \max_{\pi} \log p(\pi) + \sum_{i=1}^n \log p(x_i \mid \pi).\end{aligned}$$

- MAP is found via penalized likelihood problem

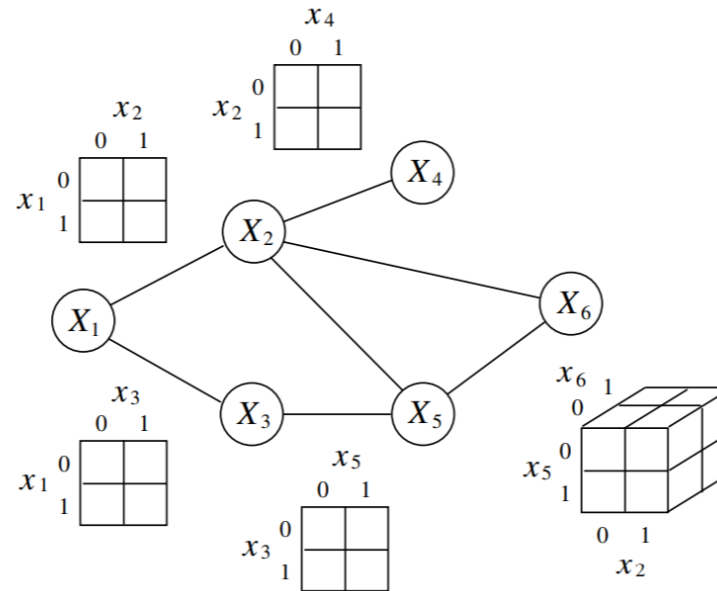
Bayesian statistics

- Bayesian statistics → treat all statistical inference as probabilistic inference
 - Even though data as observed, treat it as *random*

Assume: known structure,
unknown parameters



Generating joint distributions



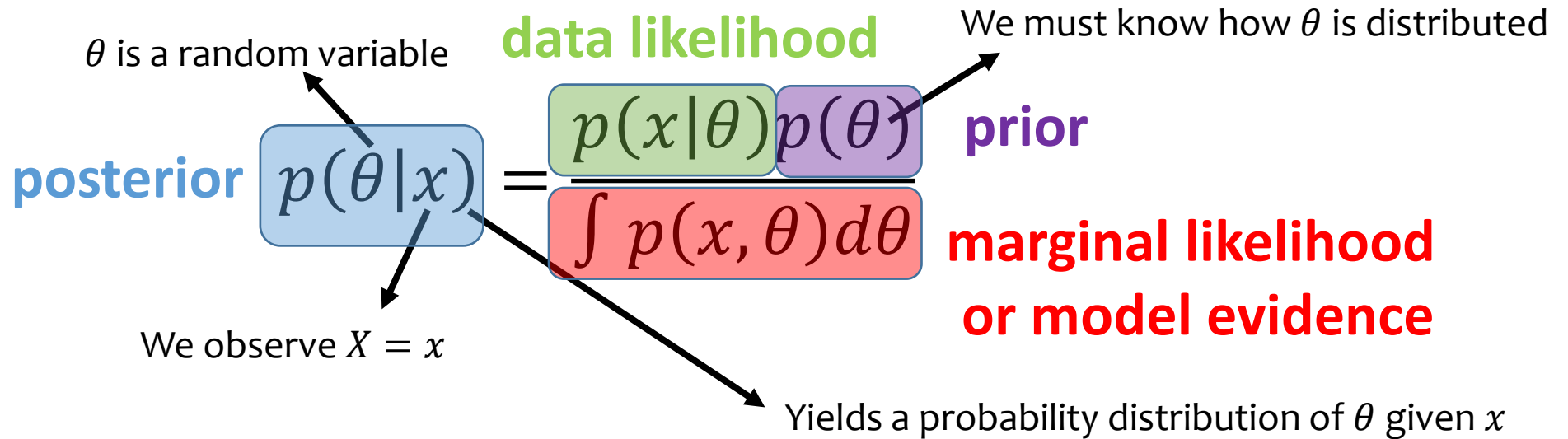
As probabilists, this conditional probability tells us how well known values of θ fit the data.

$$p(x|\theta)$$

As statisticians we need to invert this relationship.

Bayesian statistics

- As Bayesians, we reason about data likelihood in the context of Bayes theorem
 - We make several assumptions:



Frequentist statistics

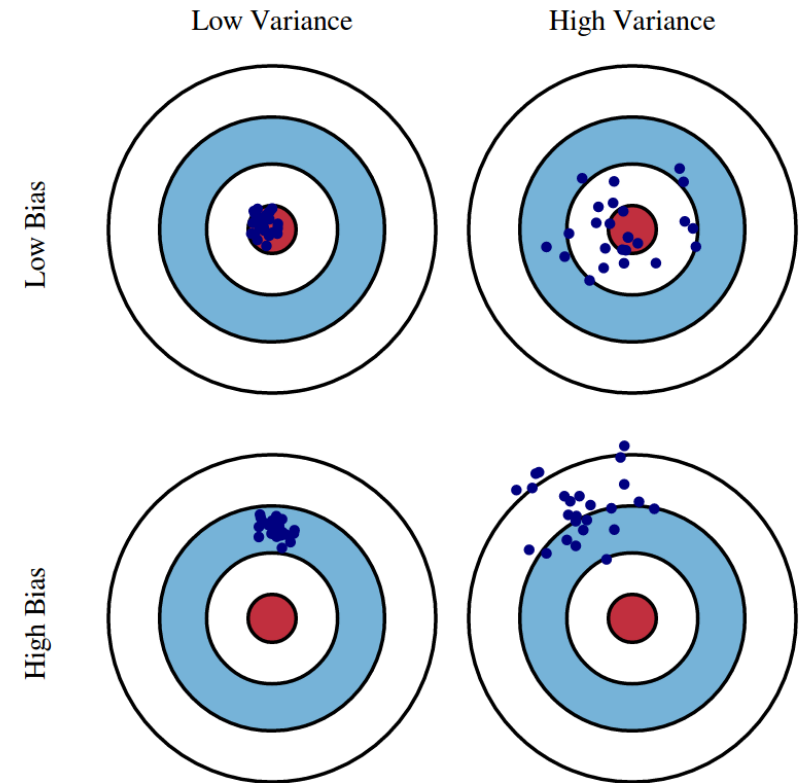
- Avoids ‘priors’, instead the goal is develop an ‘objective’ statistical theory such that several statisticians employing the methodology must draw same conclusions
- Consider coin tossing
 - outcomes $X \in \{0,1\}$
 - probability of heads $\theta \in (0,1)$

$$p(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

But, what is θ ?

Frequentist statistics

- Another frequentist interpretation is that the coin has a true probability of heads, (but the data is random) so it doesn't make sense to assign a probability distribution to it.
 - But is $p(\theta)$ modelling the physics of the coin or the experimenter's uncertainty?
- Generally, frequentists rely on estimators of θ which are functions of x and evaluate properties of such estimators



Frequentist statistics

- The most common estimator is the MLE.

$$\hat{\theta}_{\text{ML}}(x) = \arg \max_{\theta} \log p(x | \theta)$$

- $\hat{\theta}_{\text{ML}}(x)$ is a function of the data and thus is a random variable
 - How close is it to the true θ^*
 - What is its variance?
- Its distribution is governed by an unknown population distribution F

Frequentist interpretation of Bayes computations

- Both the Bayes estimator and MAP can be thought of as estimators.

$$\hat{\theta}_{\text{Bayes}}(x) = \mathbb{E}[\theta | x]$$

$$\hat{\theta}_{\text{MAP}}(x) = \arg \max_{\theta} p(\theta | x)$$

- In this context, the MAP estimate includes the log posterior (frequentist translation: penalized likelihood) and the prior (frequentist translation: regularizer)

Blurry lines – Empirical Bayes

- Consider the problem of choosing the hyperparameters α for our prior on θ
- How do we do this? Can we use the data to set the hyperparameters?

Blurry lines – Empirical Bayes

- Consider the problem of choosing the hyperparameters α for our prior on θ
- How do we do this?
 - Bayesian optimization
 - Empirical Bayes

$$\begin{aligned}\hat{\alpha}_{\text{ML}}(x) &= \arg \max_{\alpha} p(x \mid \alpha) \\ &= \arg \max_{\alpha} \int p(x \mid \theta) p(\theta \mid \alpha) d\theta.\end{aligned}$$

Model selection (Bayesian)

- Goal: $p(m | x) = \frac{p(x | m)p(m)}{p(x)}$ where $p(x | m) = \int p(x, \theta | m) d\theta$
 $= \int p(x | \theta, m) p(\theta | m) d\theta$

- Selecting a model, e.g. m^{MAP} , or fully Bayesian treatment

$$\begin{aligned} p(x_{new} | x) &= \int \int p(x_{new}, \theta, m | x) d\theta dm \\ &= \int \int p(x_{new} | \theta, m) p(\theta, m | x) d\theta dm \\ &= \int \int p(x_{new} | \theta, m) \boxed{p(\theta | x, m)} \boxed{p(m | x)} d\theta dm. \end{aligned}$$

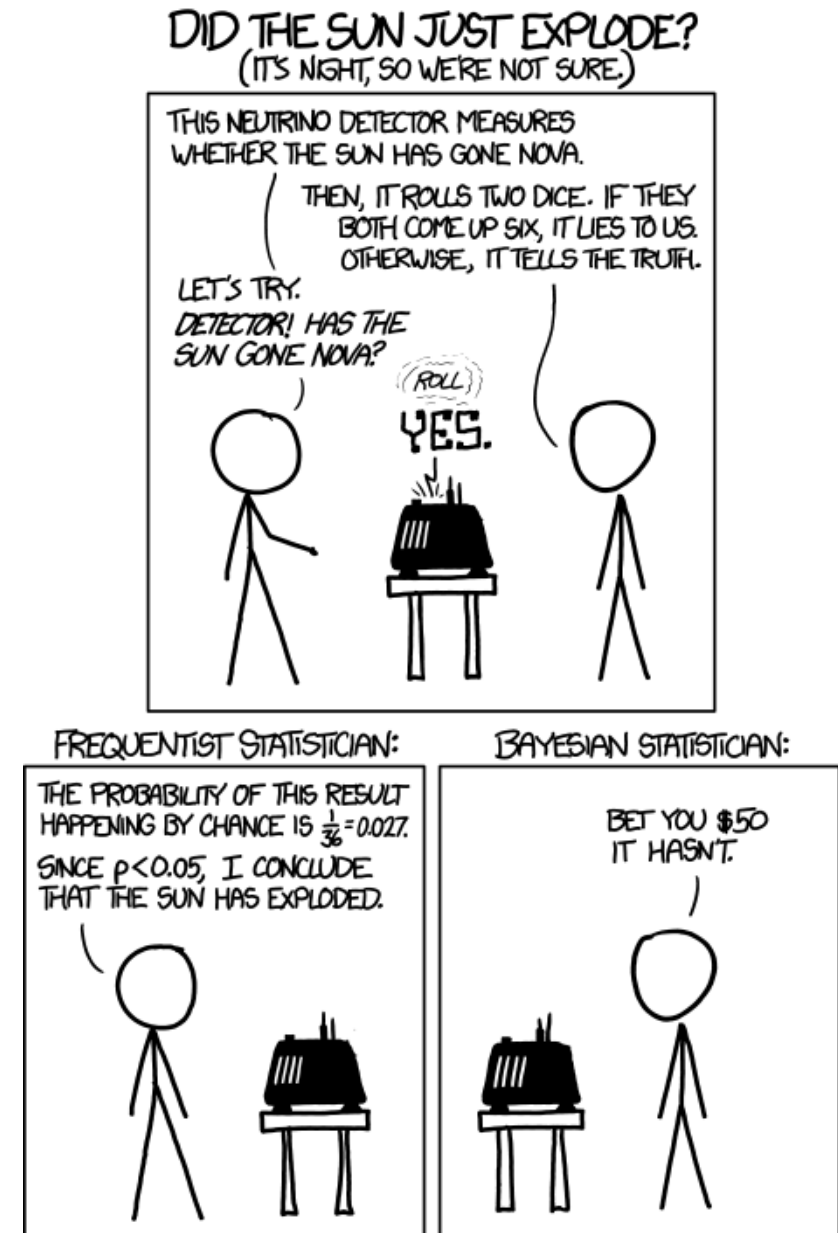
model averaging

parameter
posterior

model
posterior

Bayesian vs. Frequentist

- Both philosophies make assumptions
 - I would argue that Bayesian assumptions are more explicit
- But you can interpret Bayesian ideas in a Frequentist paradigm and vice-versa
- In Probabilistic ML we often use Bayesian computation because of its modularity and convenience, not necessarily for philosophical reasons.



Specifying a statistical model

1. Graphical model

$$p(\pi, x_{1:n}) = p(\pi) \prod_{i=1}^n p(x_i \mid \pi).$$

2. Joint distribution

$$p(\pi, x_{1:n}) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \pi^{\alpha-1} (1 - \pi)^{\beta-1} \prod_{i=1}^n \pi^{x_i} (1 - \pi)^{1-x_i}$$

3. Probabilistic generative process

Choose $\pi \sim \text{Beta}(\alpha, \beta)$

For each data point i :

Choose $x_i \mid \pi \sim \text{Bern}(\pi)$

Statistical Model Examples

Density estimation

- Density estimation problem
 - Input: observations of a random variable, X
 - Output: $\tilde{p}(\pi|X)$
- Useful for
 - Distributional classification
 - Outlier detection
 - Clustering

Density estimation example (frequentist)

- Observations X , wish to model $\tilde{p}(\pi|X)$
- What assumptions do we have to make?

Density estimation example (frequentist)

- Assume $X \sim N(\mu, \sigma^2)$: a univariate Gaussian random variable

$$p(x | \theta) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\} \quad \theta \triangleq (\mu, \sigma^2)$$

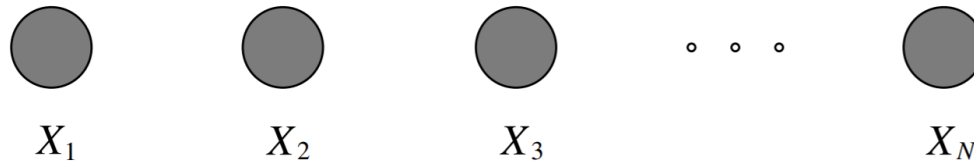
- Assume observations from N independent and identically distributed random variables

Density estimation example (frequentist)

- Assume $X \sim N(\mu, \sigma^2)$: a univariate Gaussian random variable

$$p(x | \theta) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x - \mu)^2 \right\} \quad \theta \triangleq (\mu, \sigma^2)$$

- Assume observations from N independent and identically distributed random variables



Density estimation example (frequentist)

- Joint probability of the observations is

$$\begin{aligned} p(x \mid \theta) &= \prod_{n=1}^N \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2} (x_n - \mu)^2 \right\} \\ &= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 \right\} \end{aligned}$$

- We now compute $\hat{\mu}^{MLE}$ and $\hat{\sigma}^{2MLE}$

Density estimation example (frequentist)

- $\hat{\mu}^{MLE}$

$$\begin{aligned}\frac{\partial l(\theta; x)}{\partial \mu} &= \frac{\partial}{\partial \mu} \left(-\frac{N}{2} \log(2\pi) - \frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 \right) \\ &= \frac{1}{\sigma^2} \sum_{n=1}^N (x_n - \mu).\end{aligned}$$

Density estimation example (frequentist)

- $\hat{\sigma}^{2MLE}$

$$\frac{\partial l(\theta; x)}{\partial \sigma^2} = \frac{\partial}{\partial \sigma^2} \left(-\frac{N}{2} \log(2\pi) - \frac{N}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 \right)$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu^{MLE})^2$$

Density estimation example (Bayesian)

- Estimate $p(\theta|x)$ for observed σ^2 , or $p(\mu|x, \sigma^2)$
- What should the prior density for μ be?

Conjugacy

- In our example,

$$p(\pi) \sim \text{Beta}(\alpha, \beta)$$

$$p(\pi|x) \sim \text{Beta}(\hat{\alpha}, \hat{\beta})$$

$$\hat{\alpha} = \alpha + \sum_{i=1}^n x_i \qquad \hat{\beta} = \beta + \sum_{i=1}^n (1 - x_i)$$

- The beta and Bernoulli distributions are conjugate pairs
- Most of our models will not be solved analytically, but conjugacy will be important

Density estimation example (Bayesian)

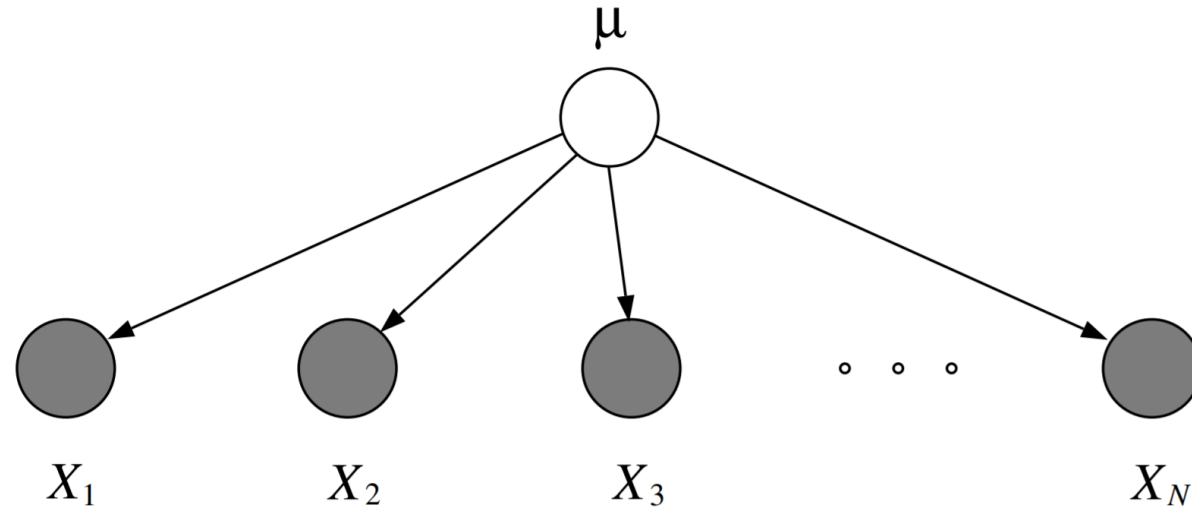
- Estimate $p(\mu|x, \sigma^2)$ for known σ^2
- What should the prior density for μ be?
 - $\mu \sim N(\mu_0, \tau^2)$
- What are μ_0, τ^2 ?

Density estimation example (Bayesian)

- Estimate $p(\mu|x, \sigma^2)$ for known σ^2
- What should the prior density for μ be?
 - $\mu \sim N(\mu_0, \tau^2)$
- Fully Bayesian approach would assign prior to μ_0 and τ^2
 - This is the hierarchical Bayesian approach
 - Diminishing returns – marginal of data typically changes less and less with more levels
- What does this graphical model look like?

Density estimation example (Bayesian)

- Data explicitly shown to be conditionally independent given μ



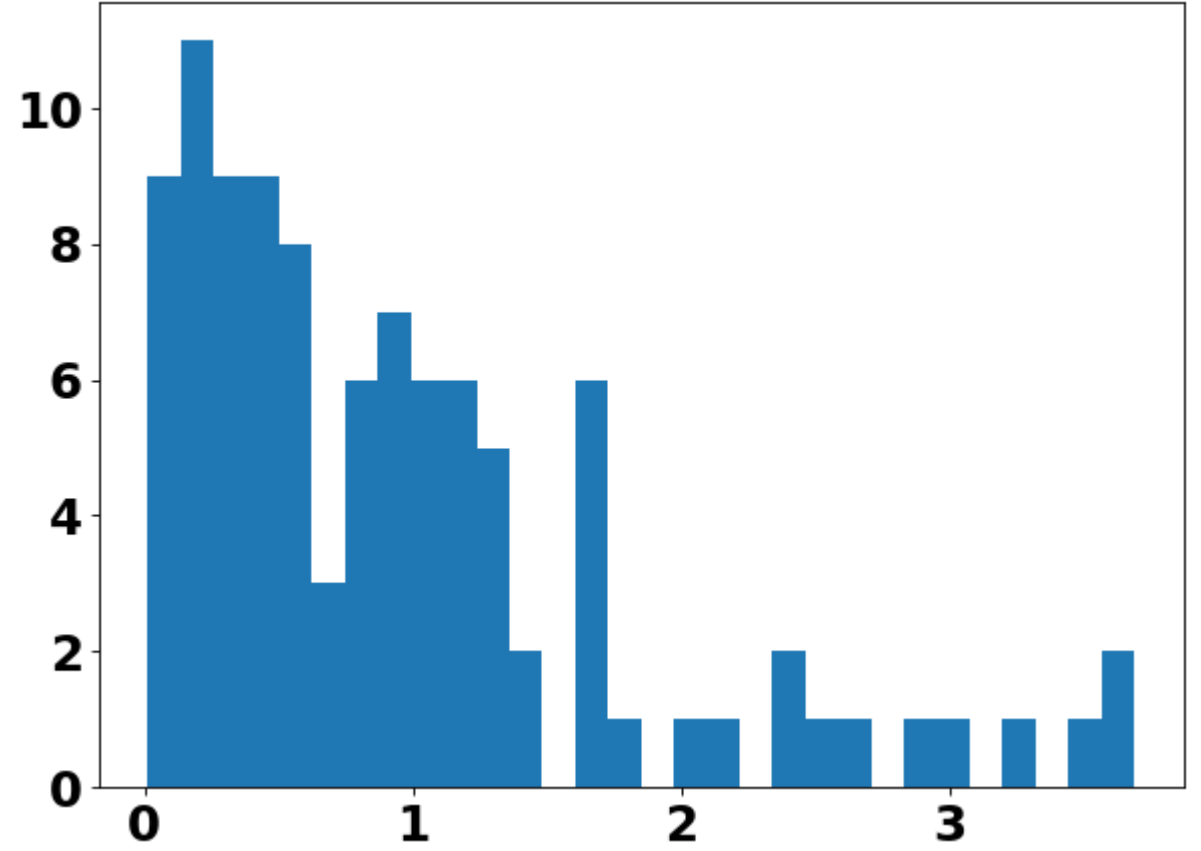
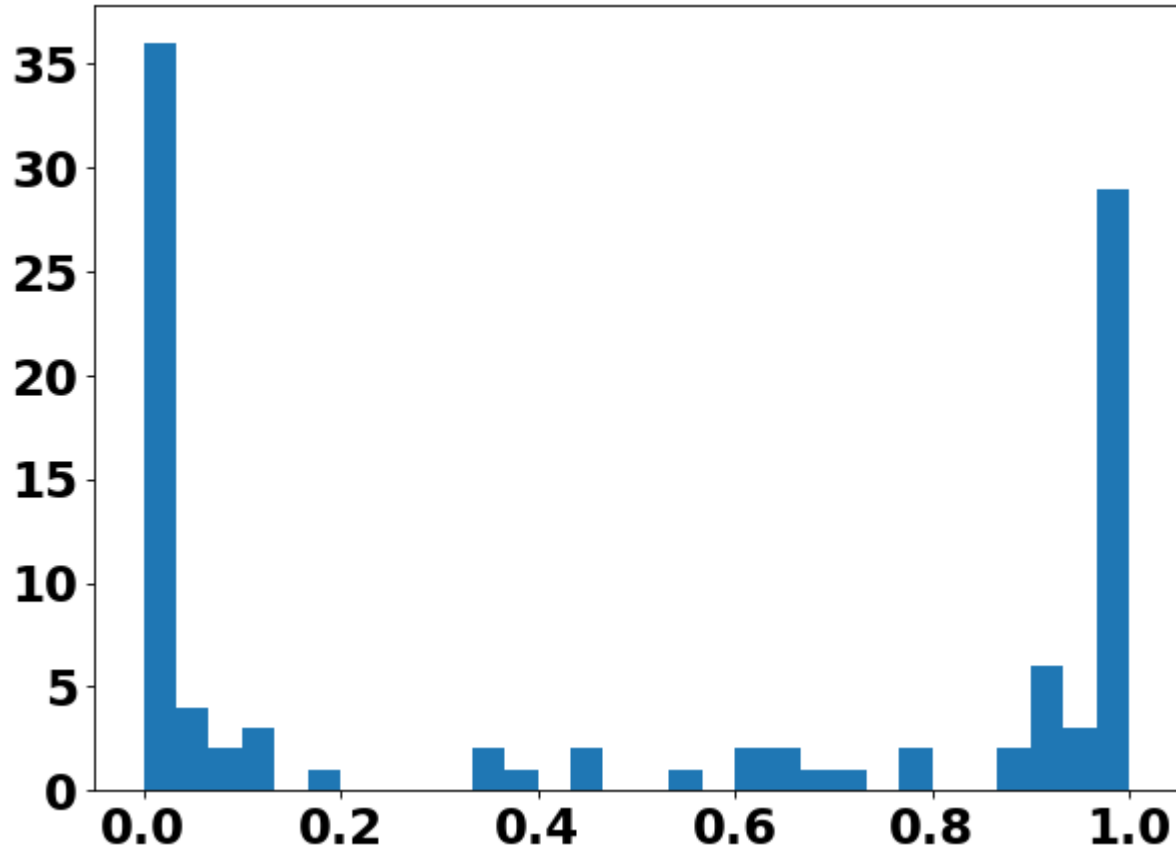
$p(x|\mu)$

$p(\mu)$

$$p(x, \mu) = \underbrace{\frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 \right\}}_{p(x|\mu)} \underbrace{\frac{1}{(2\pi\tau^2)^{1/2}} \exp \left\{ -\frac{1}{2\tau^2} (\mu - \mu_0)^2 \right\}}_{p(\mu)}$$

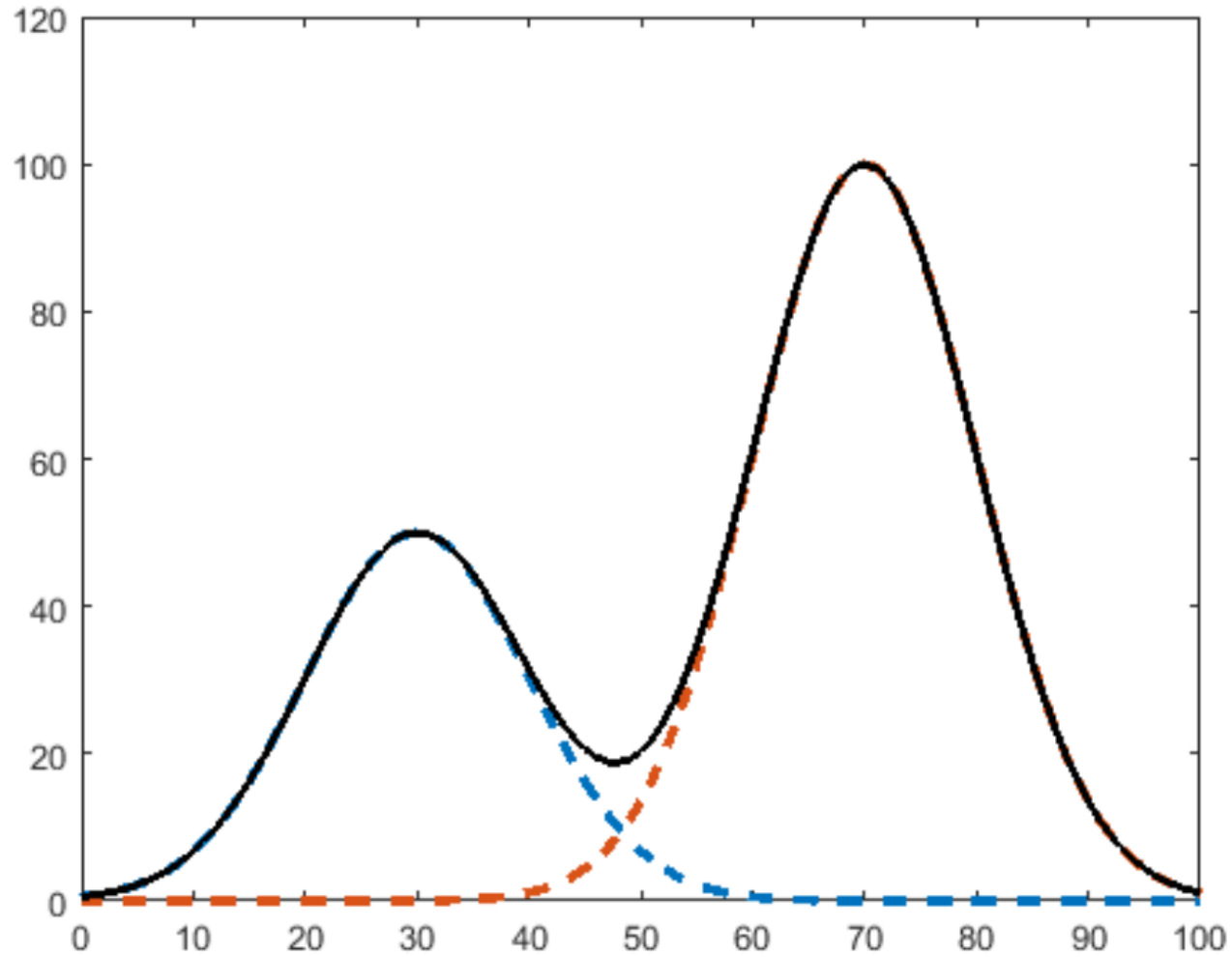
Density estimation

Normal/multinomial might not always be the best choice

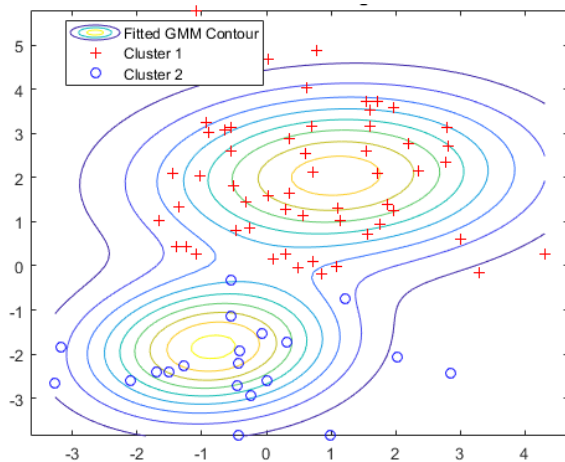
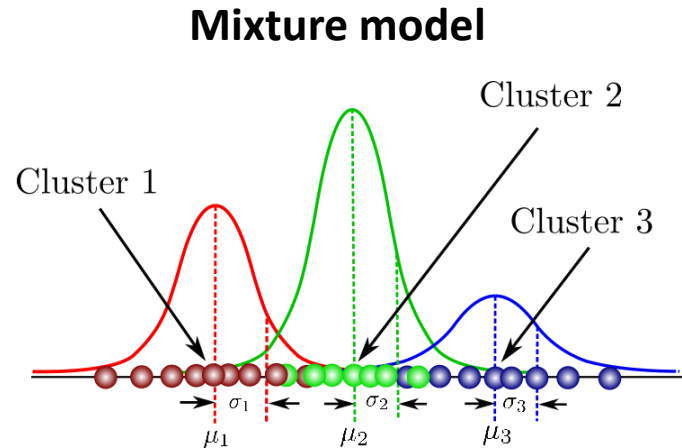


How would you model these densities? What assumptions would you make?

Density estimation



Mixture models



- We need a more expressive family of probability distributions to model multi-modal data.
- Mixture models capture subpopulation or cluster structures in sampled populations
- For example, x = Ph.D. student heights
 - The likelihood term for a single density would be low (**why?**)
 - What if we broke down the density and modelled each cluster separately?

Mixture models (frequentist)

- Assume we know the number of components K
- Given each component is distribution $f_k(x|\theta_k)$

$$p(x \mid \theta) =$$

Mixture models (frequentist)

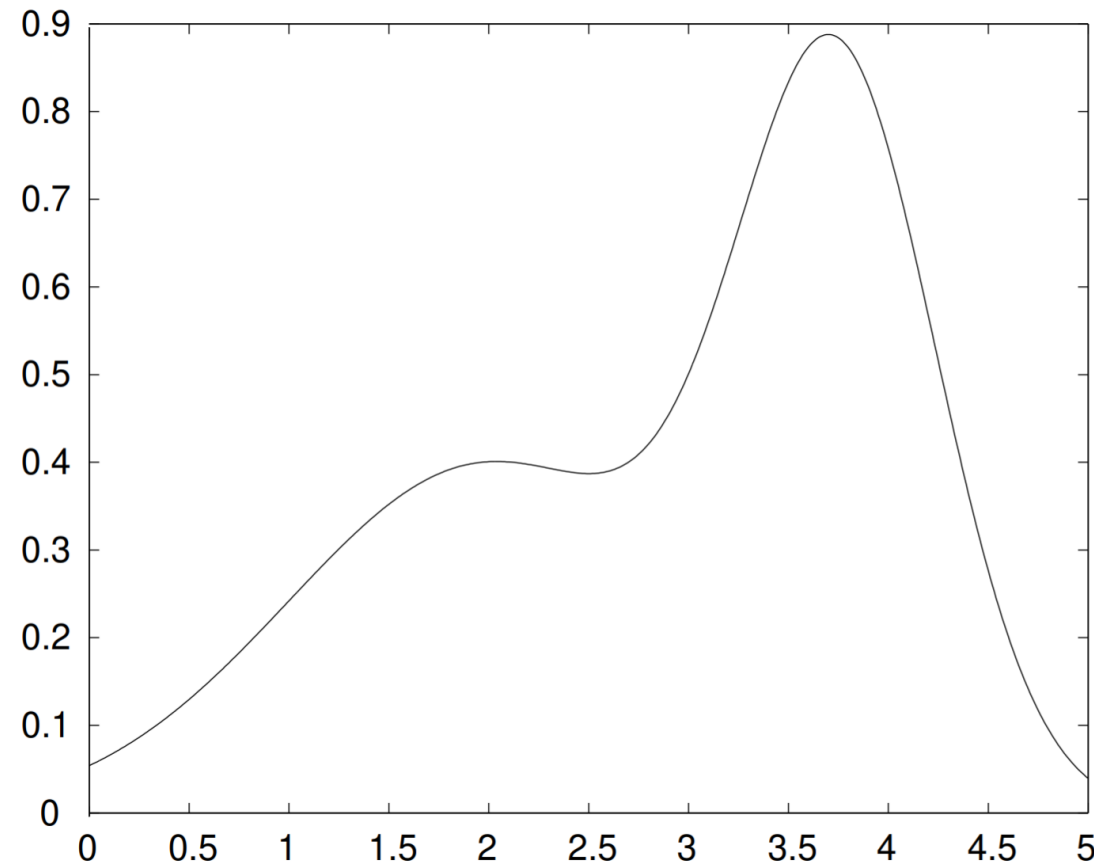
- Assume we know the number of components K
- We can define the mixture density as the convex sum over component densities

$$p(x \mid \theta) = \sum_{k=1}^K \alpha_k f_k(x \mid \theta_k)$$


mixture proportions

mixture components

Example with 2 components



$$p(x | \theta) = \alpha_1 \mathcal{N}(x | \mu_1, \sigma_1^2) + \alpha_2 \mathcal{N}(x | \mu_2, \sigma_2^2)$$

Bayesian treatment for mixture models

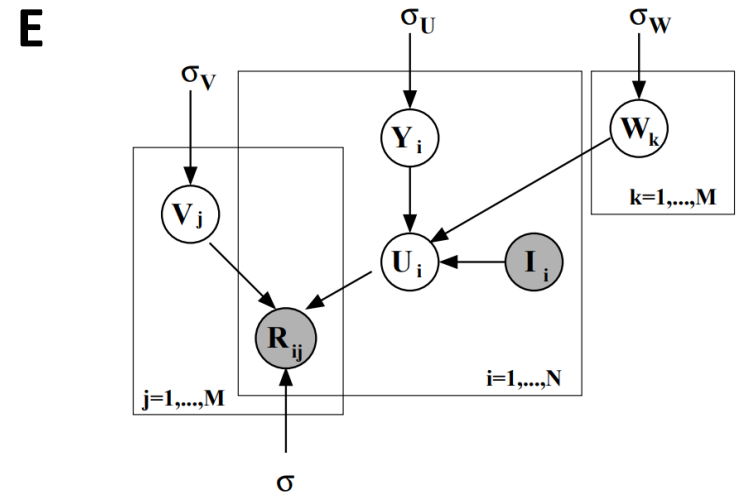
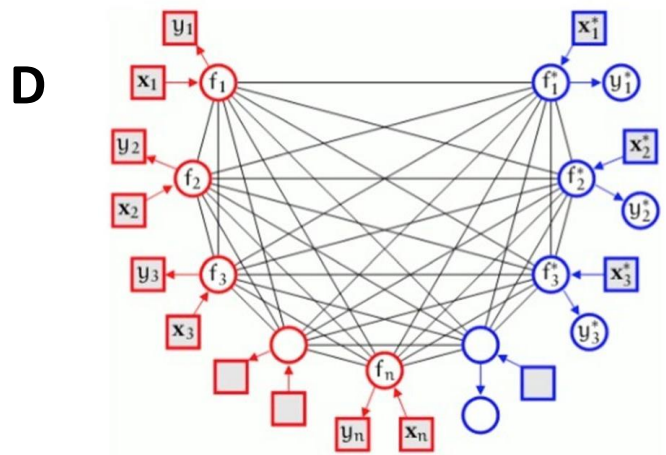
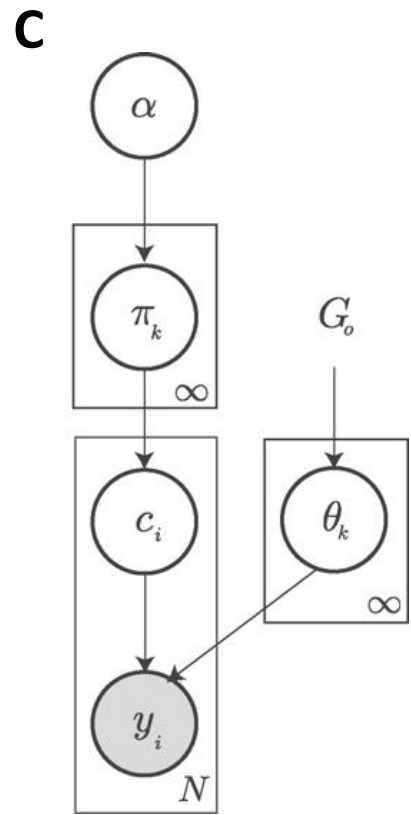
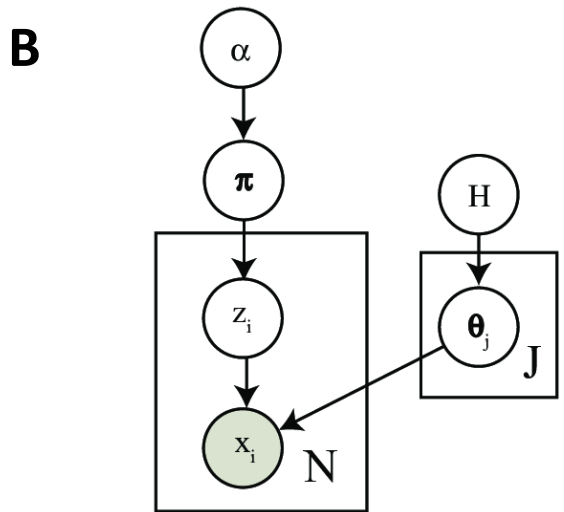
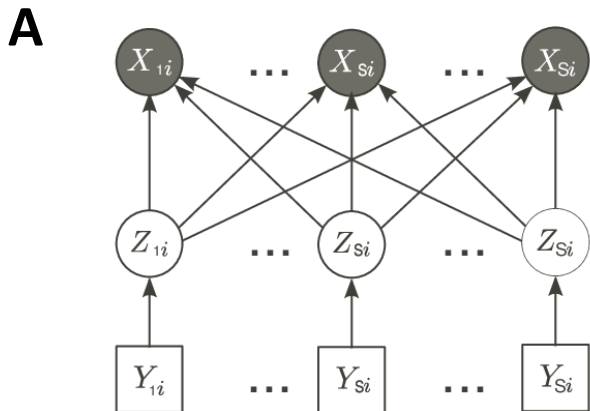
- What are the hidden random variables?

Bayesian treatment for mixture models

- What are the hidden random variables?
 - cluster assignments for each point
 - mixture proportions
 - cluster parameters for each cluster (mixture components)
- Model the cluster assignments as a categorical



Which of the following are mixture models?



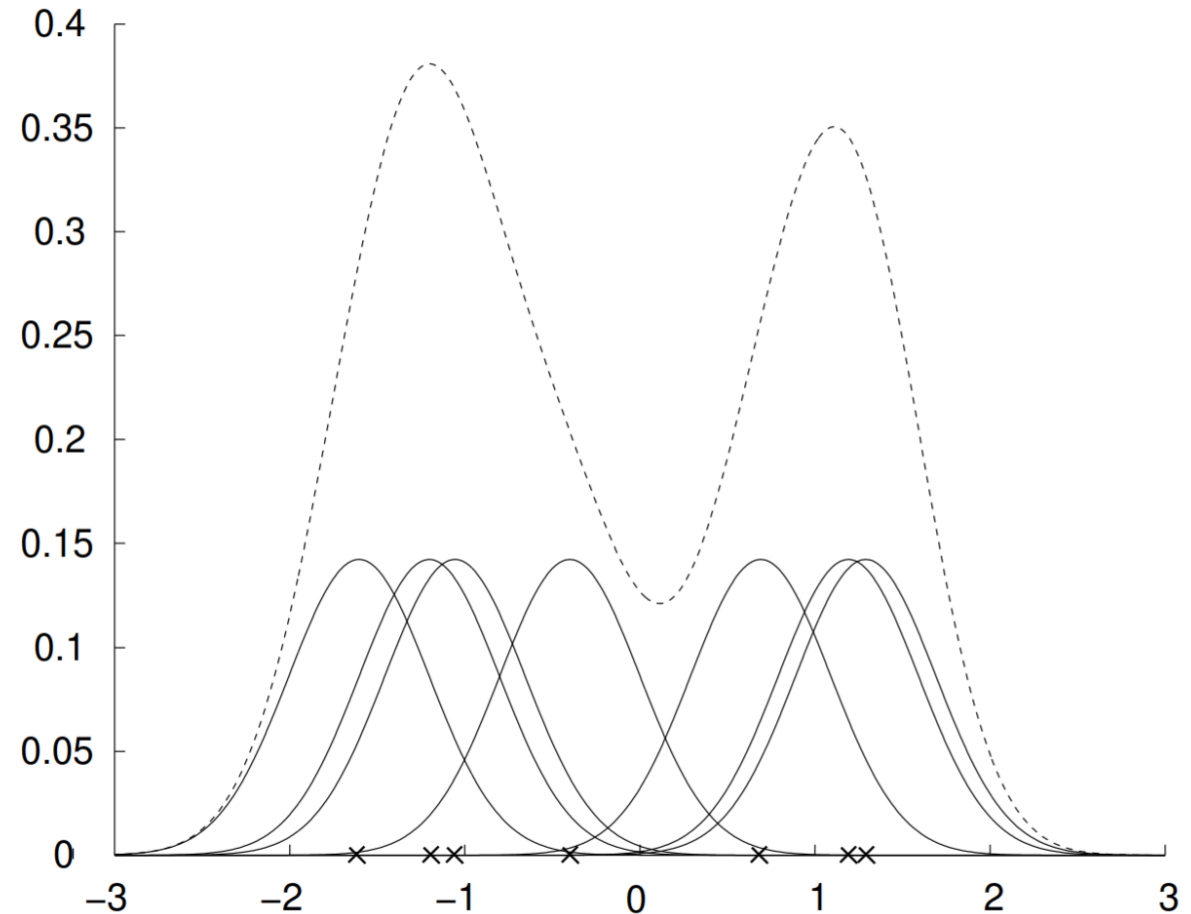
Nonparametric density estimation

- What if it is not obvious how to partition the data or how to categorize the mixture components?
- Nonparametric density estimation enables mixture modeling in knowledge-poor scenarios
- Kernel density estimation
 - Each data points represents a small bit of probability mass
 - Assume the density is smooth, superimpose smaller densities to create overall density

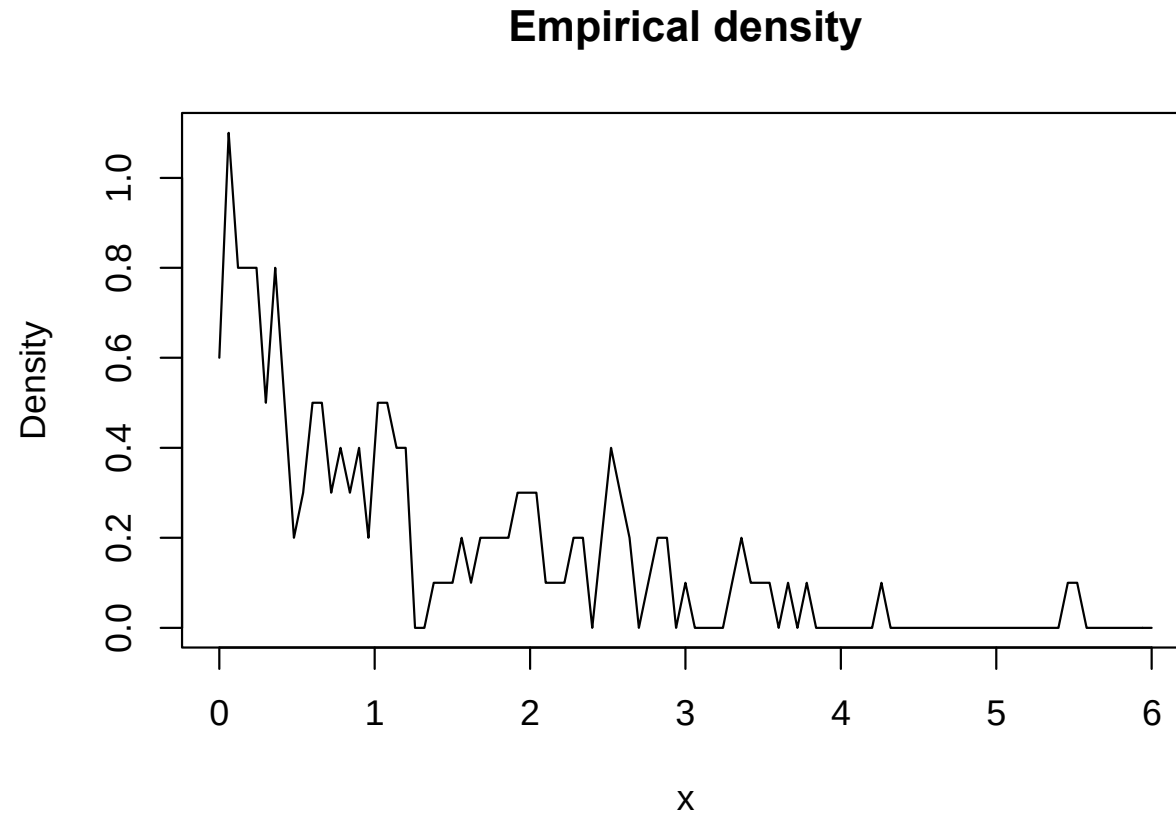
Kernel density estimation

- Let $k(x, x_n, \lambda)$ be a kernel function
 - Non negative, integrates to 1 w.r.t. x
 - x_n determines the location
 - λ is the bandwidth determining the 'smoothness'
- Superimposing N kernels, we obtain the density

$$\hat{p}(x) = \frac{1}{N} \sum_{n=1}^N k(x, x_n, \lambda)$$



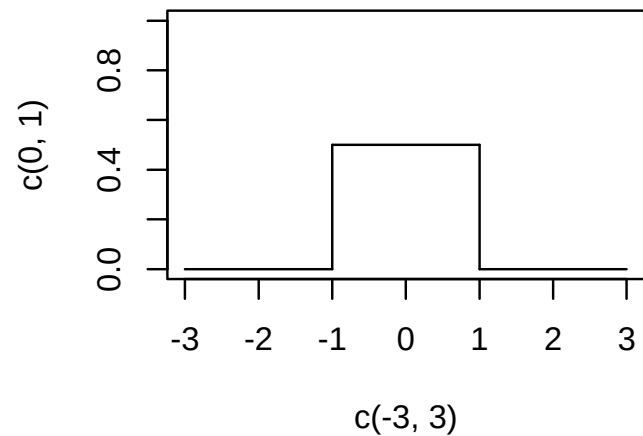
The exponential sample



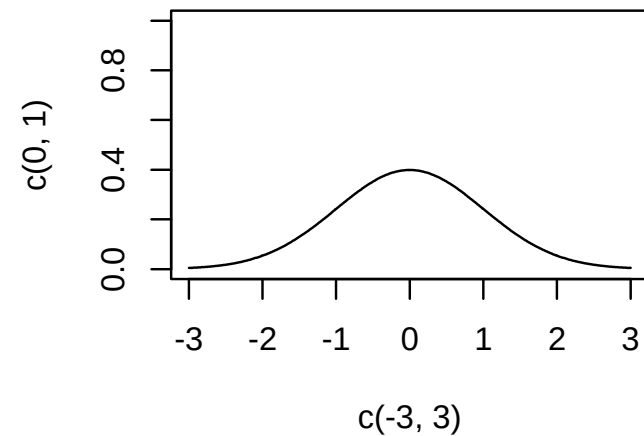
Kernels

bandwidth $\lambda = 1$

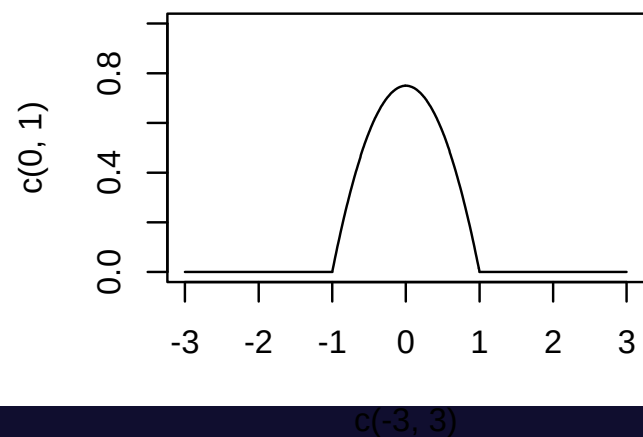
$K(x) = \frac{1}{2} 1(|x| \leq 1)$
Rectangular



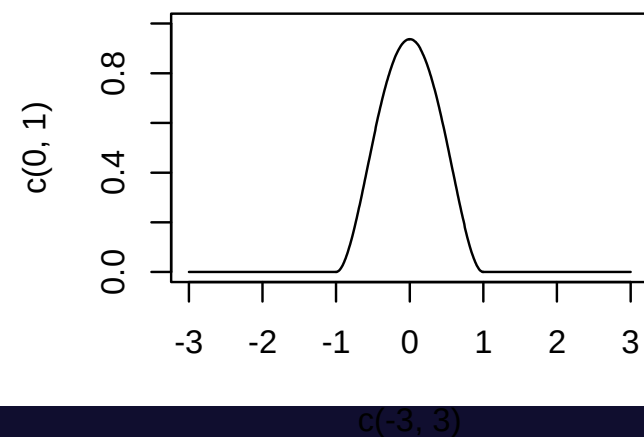
$K(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2 / 2)$
Gaussian



$K(x) = \frac{3}{4} (1 - x^2), |x| \leq 1$
Epanechnikov



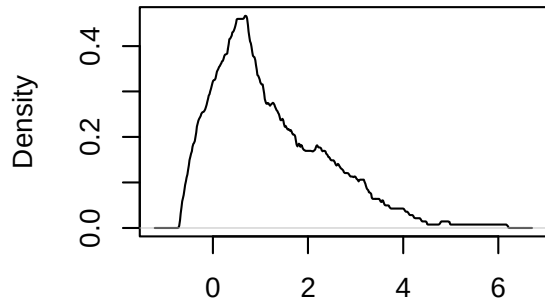
$K(x) = (1 - x^2)^2, |x| \leq 1$
Biweight



The exponential sample

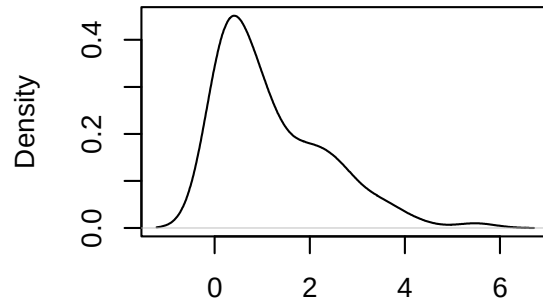
Different kernels

Rectangular



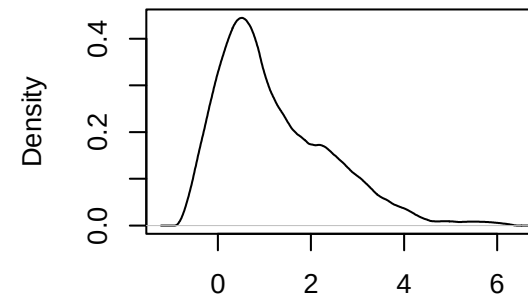
N = 100 Bandwidth = 0.4082

Gaussian



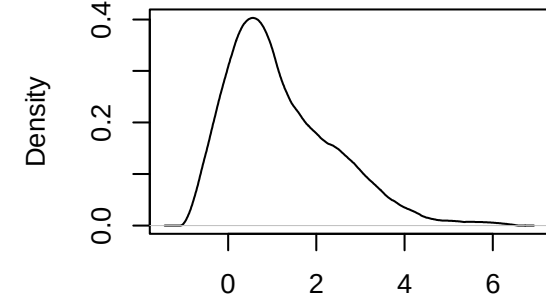
N = 100 Bandwidth = 0.4082

Silverman's rule of thumb



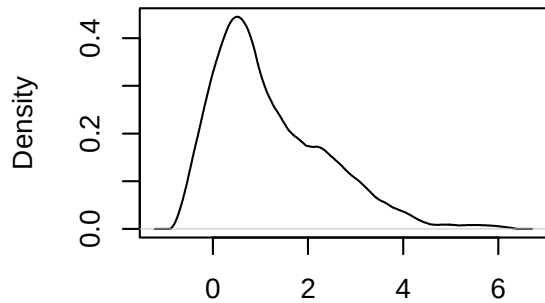
N = 100 Bandwidth = 0.4082

Scott's rule of thumb



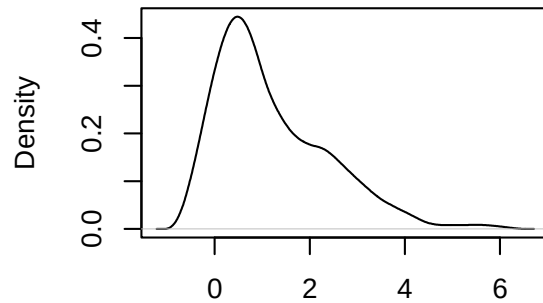
N = 100 Bandwidth = 0.4807

Epanechnikov



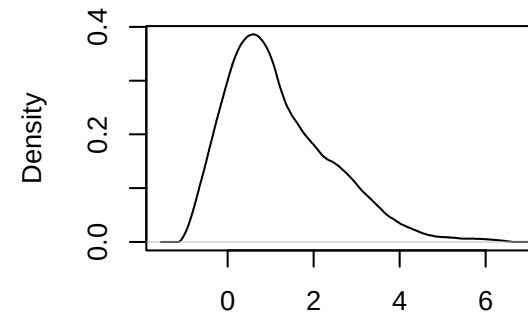
N = 100 Bandwidth = 0.4082

Biweight



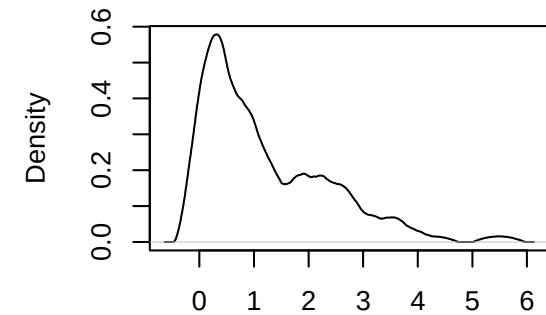
N = 100 Bandwidth = 0.4082

Biased crossvalidation



N = 100 Bandwidth = 0.5171

Sheather and Jones



N = 100 Bandwidth = 0.2131

Parametric vs. Nonparametric

- Parametric
 - As N increases...
- Nonparametric
 - As N increase...

How does the 'complexity' of the mixture distribution change?

Parametric vs. Nonparametric

- Parametric

- As N increases, the complexity of the family of probability distributions stays the same
- As N increases...

- Nonparametric

- As N increase, the complexity increases
- As N increases...

How does the number of parameters in mixture distribution change?

Parametric vs. Nonparametric

- Parametric

- As N increases, the complexity of the family of probability distributions stays the same
- As N increases, the number of parameters stays the same
- Must “select” the right model (what is k ? What is the parametric form?)
- Faster to converge

- Nonparametric

- As N increase, the complexity increases
- As N increases, the number of parameters increases
- Don't have to set the distributional form or the number of components
- Slow to converge

Regression

- Model dependence between a covariate X and response Y , $P(\mathbf{y}|\mathbf{x})$
- Use Bayes Theorem?

$$y_n = \beta^T x_n + \epsilon_n$$

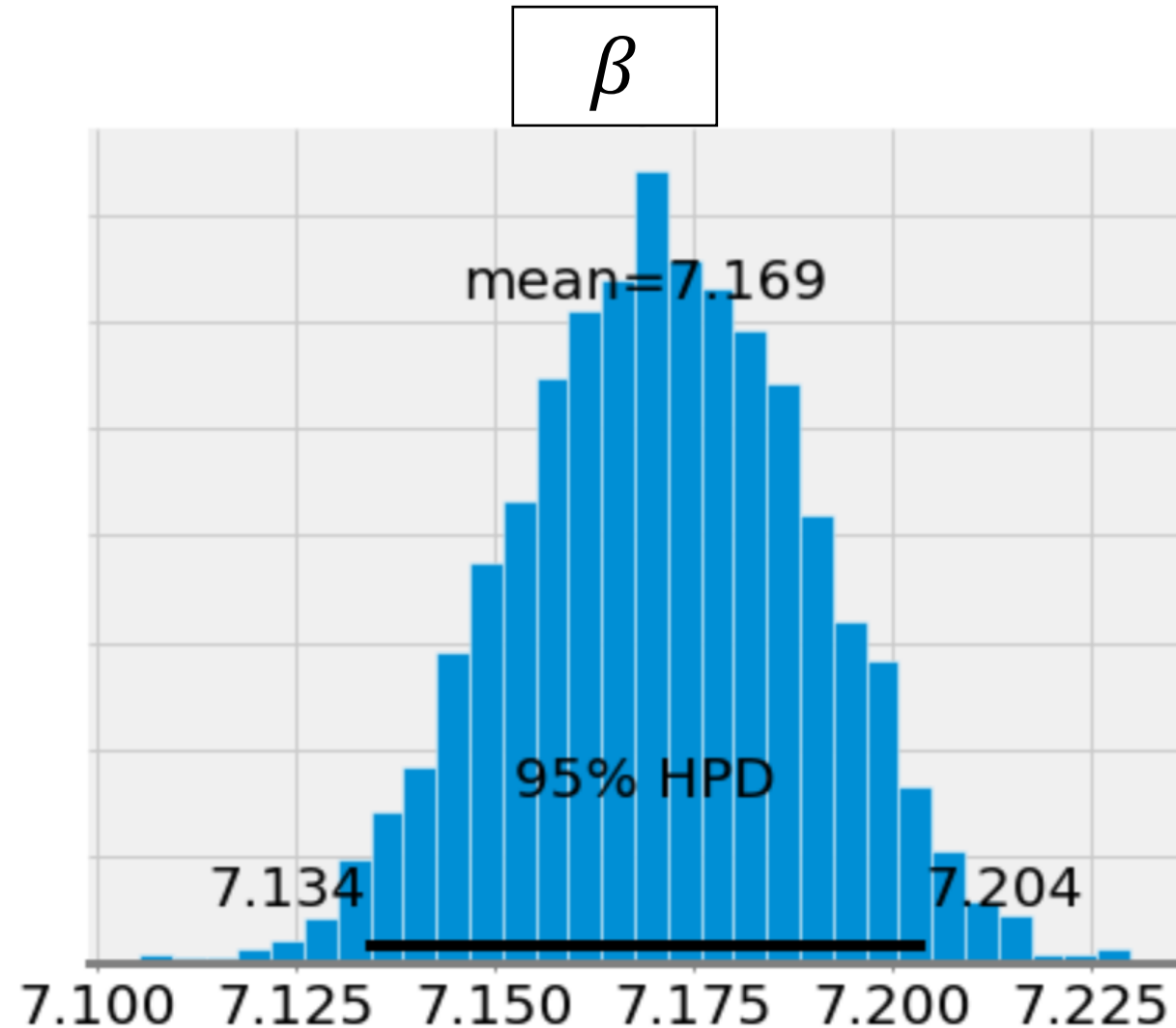
- Frequentist: learn β that maximizes $l(\beta; x) = \log p(\mathbf{y}|\mathbf{x}, \beta)$
- Bayesian: compute posterior $P(\beta|\mathbf{x}, \mathbf{y})$

Regression (Bayesian)

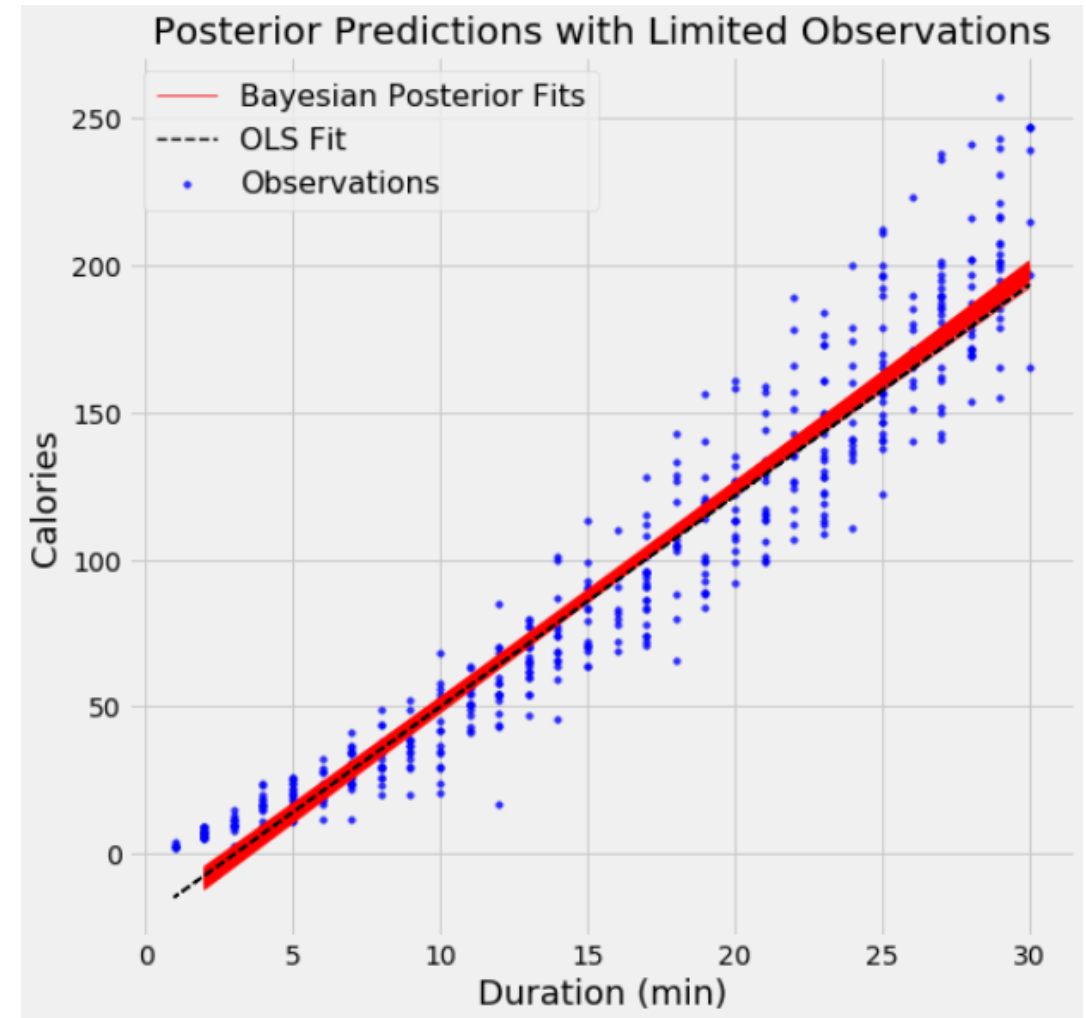
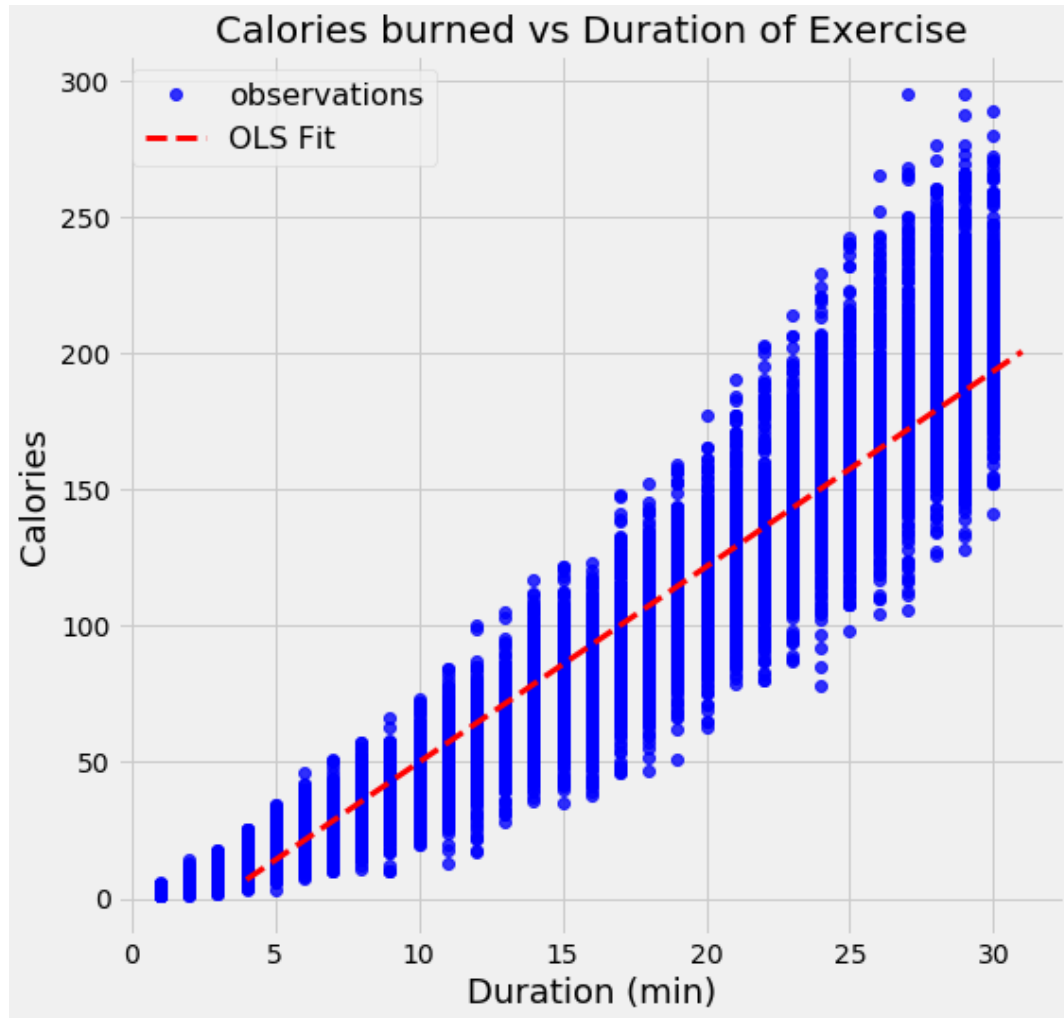
$$Y_n = \beta^T x_n + \epsilon_n$$

- In Bayesian linear regression, we wish to compute the posterior.

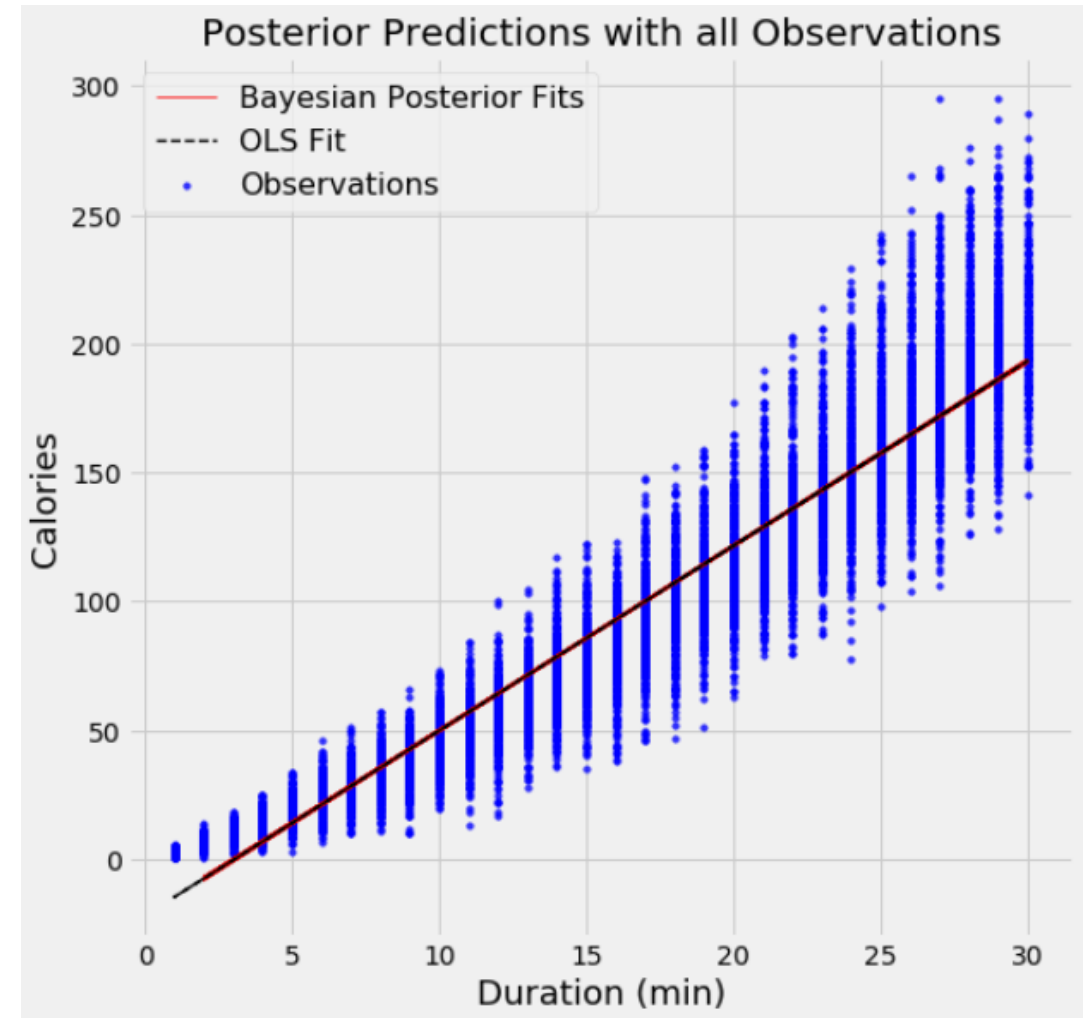
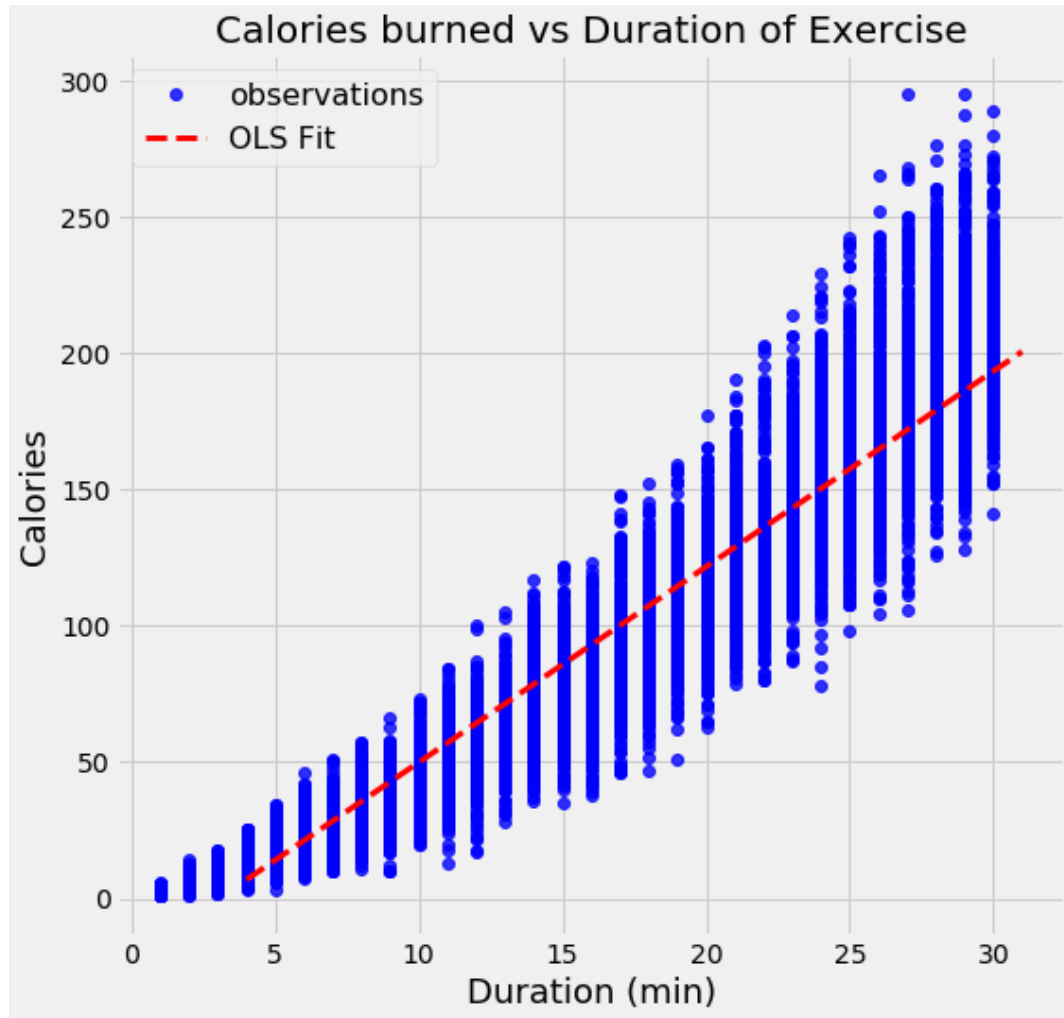
$$P(\beta|x, Y) = \frac{P(Y|x, \beta)P(\beta|x)}{P(Y|x)}$$



OLS vs. Bayesian Regression limited data



OLS vs. Bayesian Regression full data



Generative model?

- Is regression a generative model?

Generative model?

- Is regression a generative model?
- No, it is a **discriminative** model.
 - Each output is conditioned on inputs that are not modelled

Training vs. testing

- In density estimation, did we have both the inputs and response?

Training vs. testing

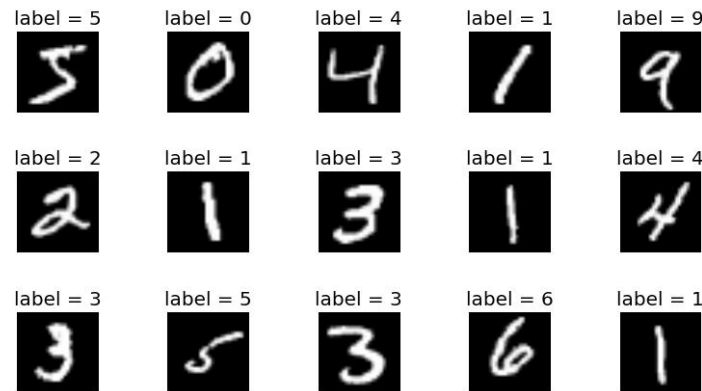
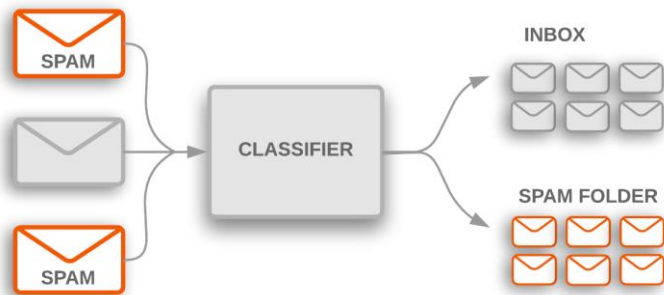
- In density estimation, did we have both the inputs and response?
 - We never observe the hidden variables.
- In regression, did we have both the inputs and response?

Training vs. testing

- In density estimation, did we have both the inputs and response?
 - We never observe the hidden variables.
- In regression, did we have both the inputs and response?
 - We fit a model on observed (training) data, then acquire more input data (test) to predict the response (hidden in the test data).

Classification

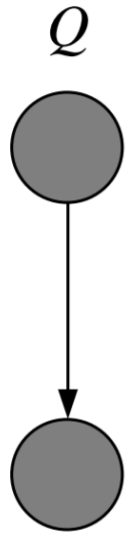
- Similar to regression, but response variable ranges over a finite set
 - $X = \text{feature vector}$
 - $Q = \text{class label}$
- Input: X
- Output: For each $x_i \in X$, a class label $q_i \in Q$



individuals	genome variants	Disease status
	CGACCCTCCCGGA	→ +
	CGACCCTCCCGGA	→ +
	CGACACTATCAGA	→ +
	CGACACTATCAGA	
	AACTCTGATGGGA	•
	AACTCTGATGGGA	•
	AACTCTGATGAGA	•
	AACTCTGATGAGA	
	AACTCTGATGGAG	→ -
	AACTCTGATGGGA	→ -

Classification (two approaches)

Generative



- Imbue variables with density $p(x|q)$ and class priors $p(q)$
- HMMs, mixture models, Naïve Bayes, LDA, etc...

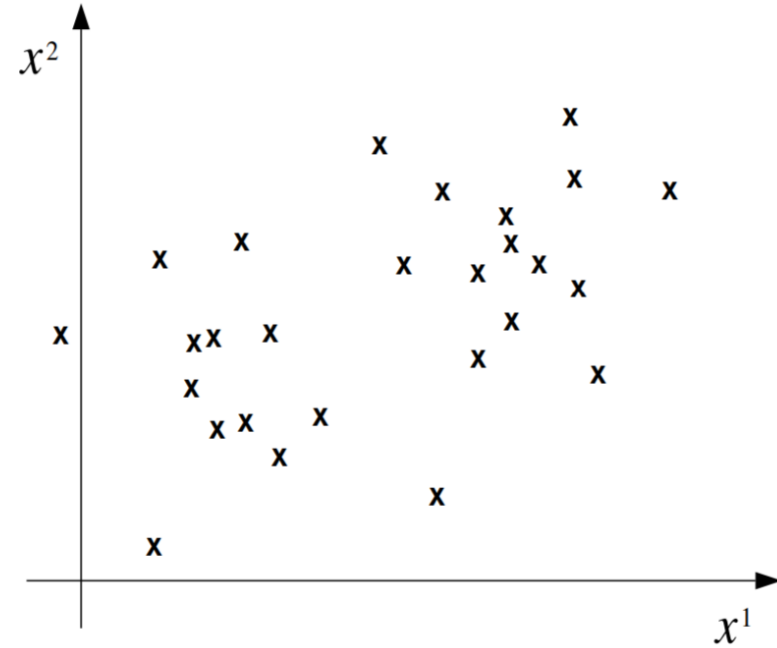
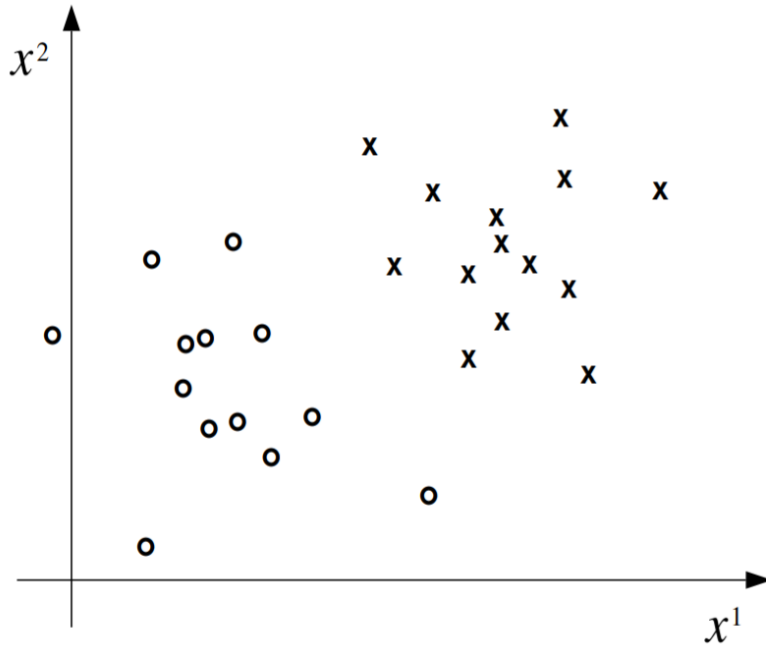
Discriminative



- Determine class membership for data point x with $\arg \max_q p(q|x)$
- E.g. logistic regression, SVM, neural network, etc..
 - Differentiate classes without explicitly modelling them

Classification?

- What if we don't know the class labels?



Clustering

- Basic graphical model the same as classification except...



Factor models

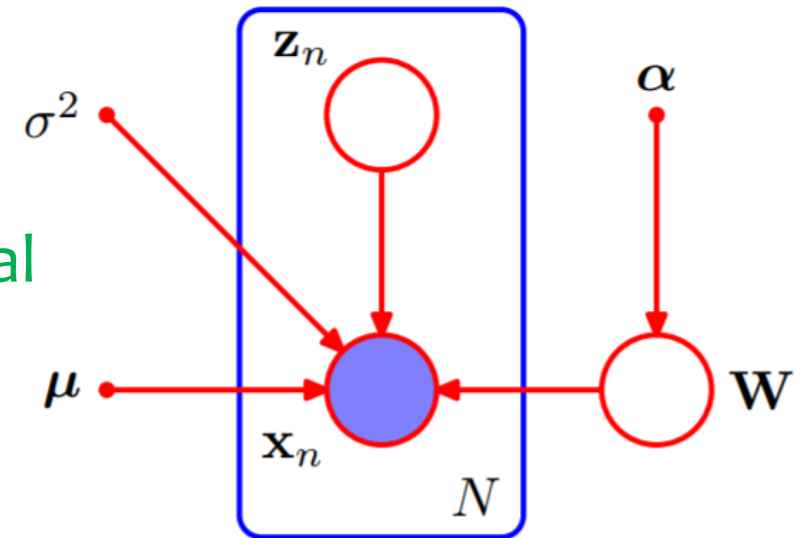
- Embed high-dimensional data in a low-dimensional space
- Data is a linear function of low dimensional latent coordinates (plus Gaussian noise)

$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \boldsymbol{\Psi})$$

factor loading

mean matrix

diagonal



Other models...

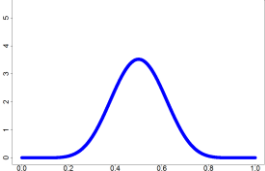
- Bayesian nonparametric models
 - Gaussian processes
 - Hierarchical Dirichlet processes
 - Indian buffet processes
- Factorial models
- Mixed-effect models
- Hierarchical models
- State-space models
- *probabilistic modeling is not a “cookbook”*

BML workflow (Box's loop)

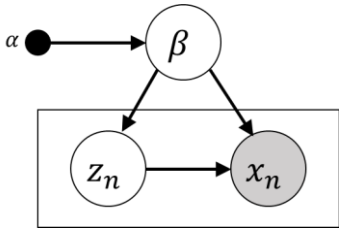
domain expertise



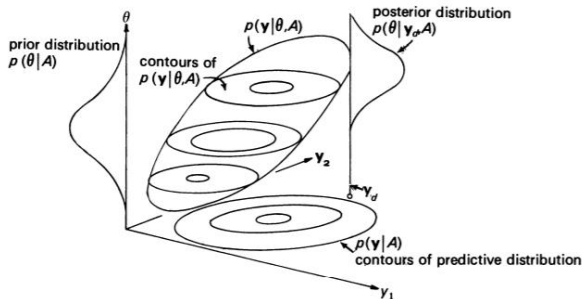
prior knowledge



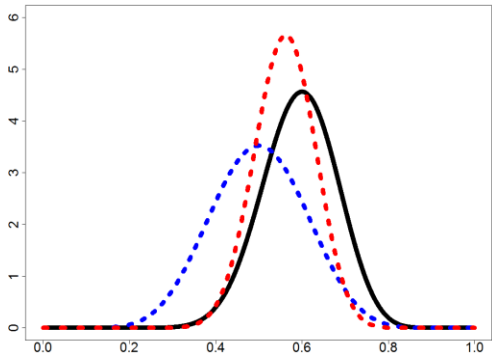
Probabilistic modeling



model criticism



Posterior inference



predictions & exploration

CGACCCTCCCAGTGGAGCCAG
CGACCCTCCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG

data

CGACCCTCCCAGTGGAGCCAG
CGACCCTCCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG

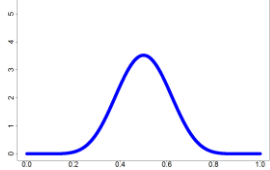
Box's loop

1

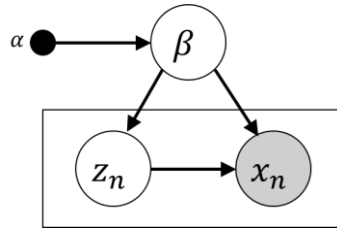
domain expertise



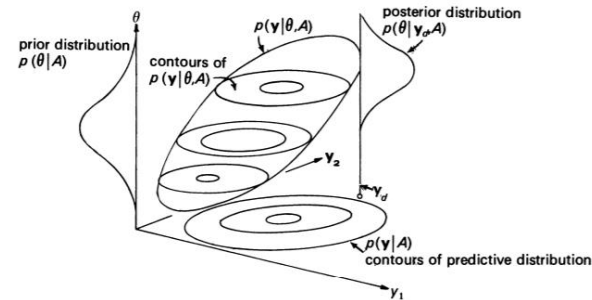
prior knowledge



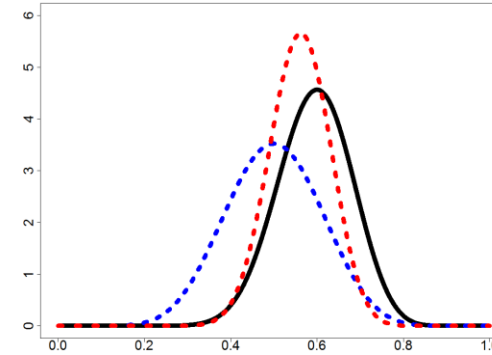
Probabilistic modeling



model criticism



Posterior inference



predictions & exploration

CGACCTCCAGTGGAGCCAG
CGACCTCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAAATGGAGCCAG
AACTCTGATGAAATGGAGCCAG
AACTCTGATGATAGAGCCAG
AACTCTGATGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG

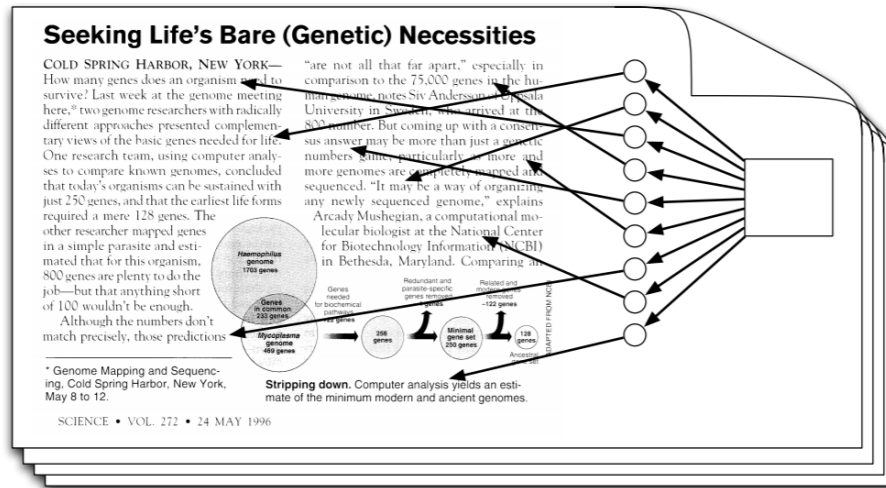
data

CGACCCTCCCAGTGGAGCCAG
CGACCCTCCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAACAG

Building models

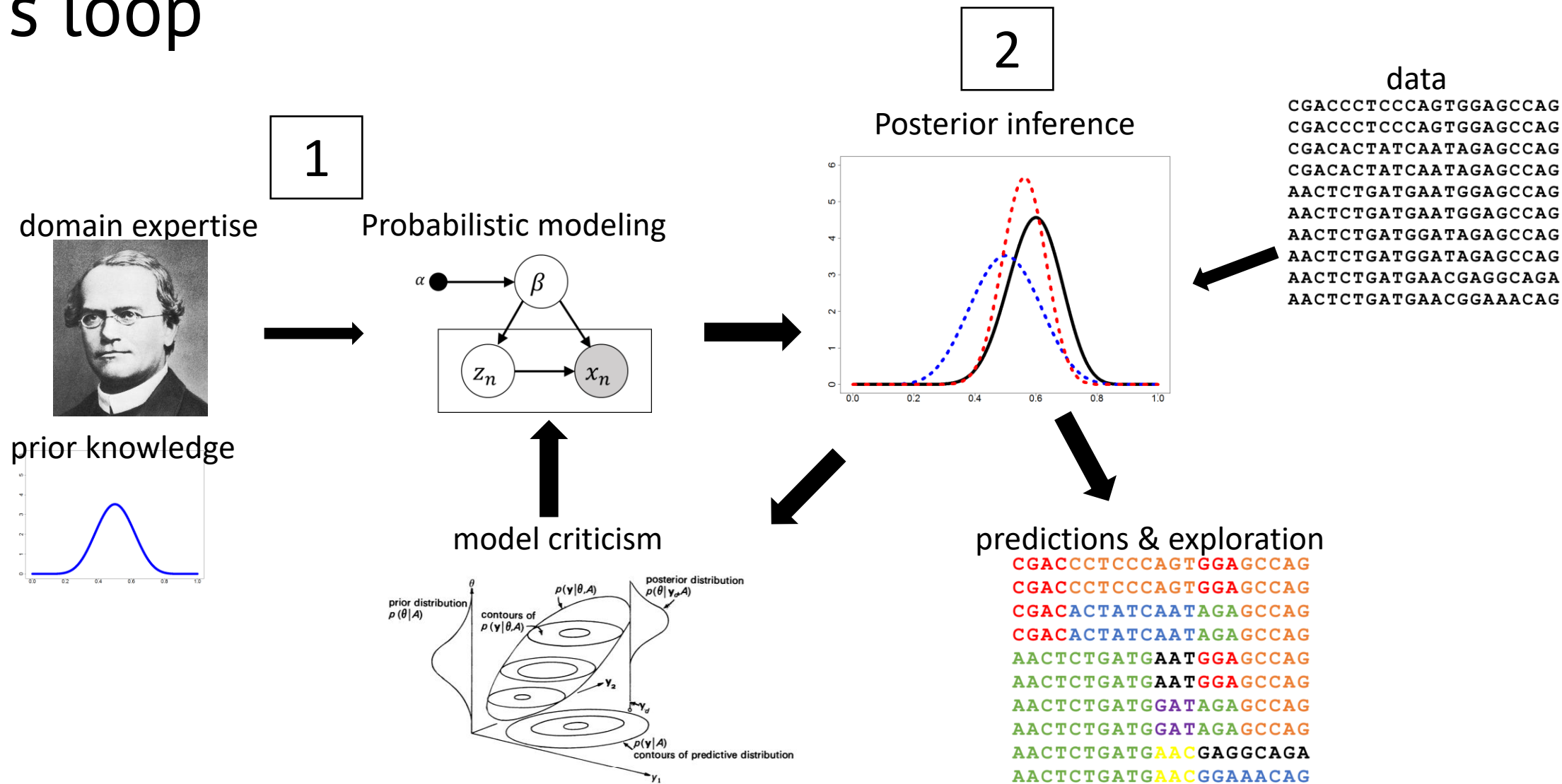
1. Set up probability models

- Write down multiple models, in increasing order of complexity
- Joint distribution of all observed and hidden random variables
- Use your domain expertise and assumptions about data generation
- Think about what quantities you want to infer from this model



- Model 0: document = bag of words from abstract
- Model 1: document = bag of words from full text
- Model 2: document = hidden Markov model
- Model 3: document = variable length HMM

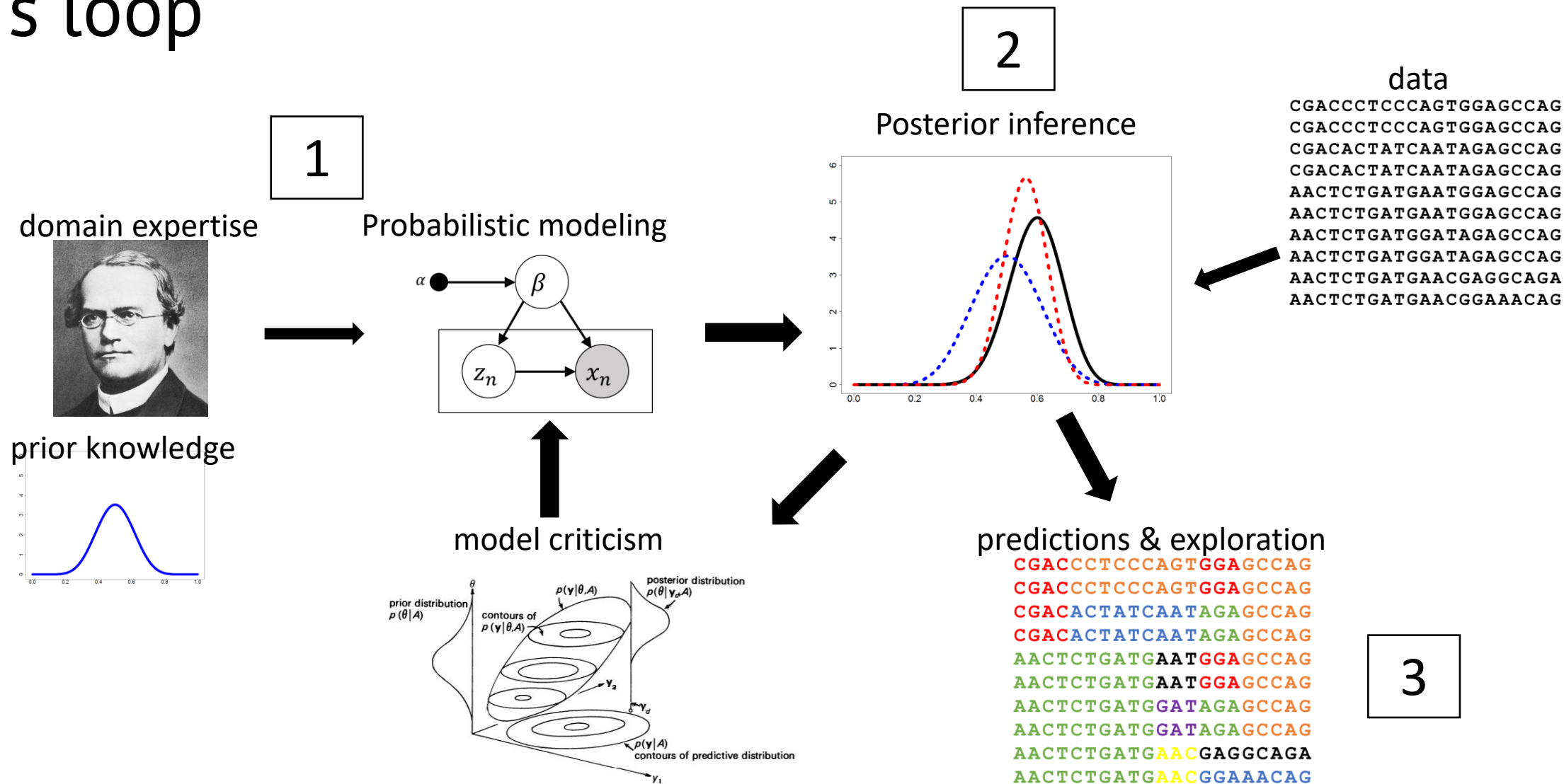
Box's loop



Building models

2. Conditioned on observations, calculate and interpret the posterior
 - How does the likelihood affect the posterior?
 - The prior?

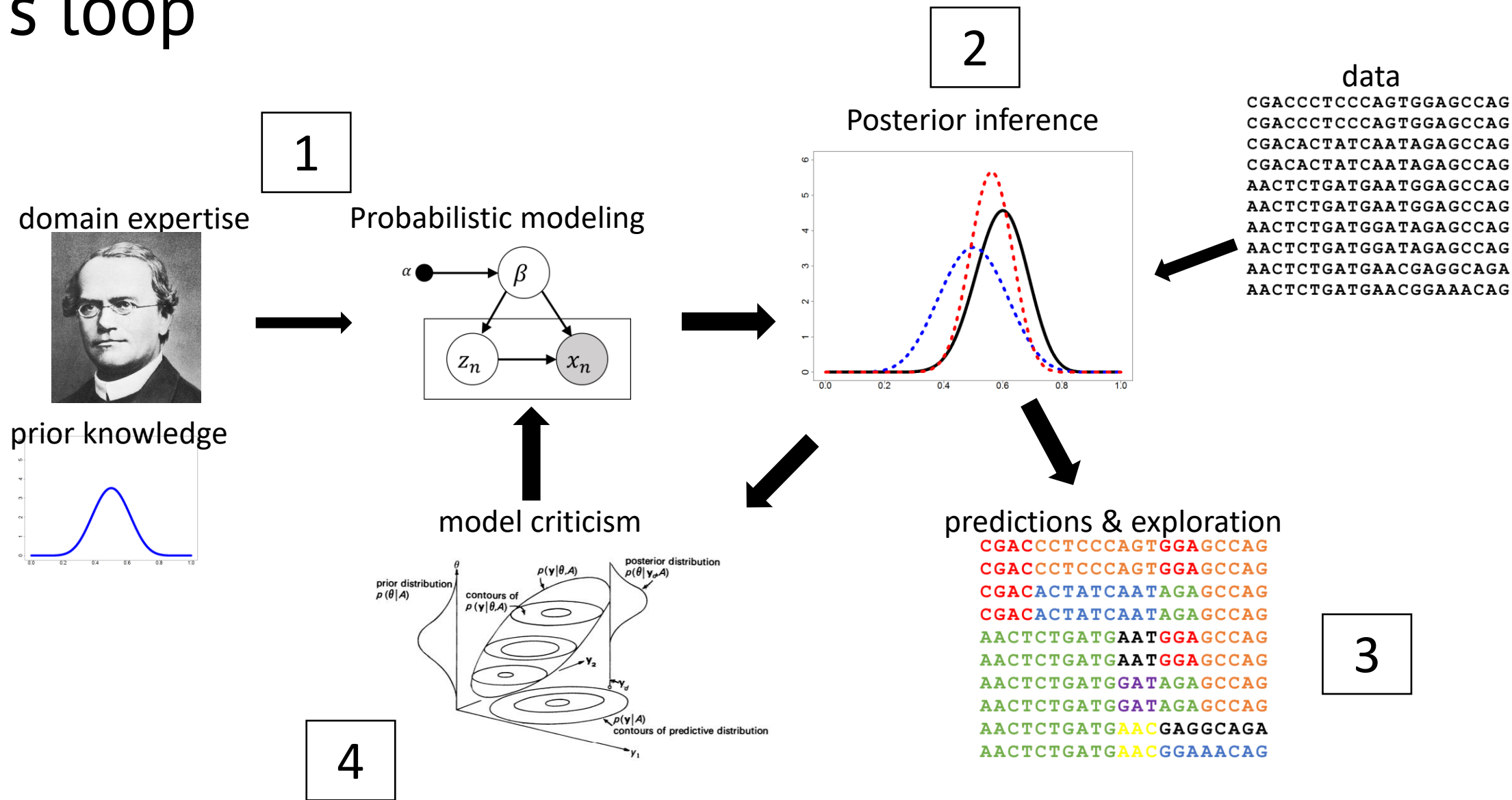
Box's loop



Building models

2. Conditioned on observations, calculate and interpret the posterior
 - How does the likelihood affect the posterior?
 - The prior?
3. Evaluate the model fit and posterior implications
 - Does the model fit the data?
 - Are the conclusions reasonable?
 - How sensitive are results to perturbation of model?

Box's loop



Building models

2. Conditioned on observations, calculate and interpret the posterior
 - How does the likelihood affect the posterior?
 - The prior?
3. Evaluate the model fit and posterior implications
 - Does the model fit the data?
 - Are the conclusions reasonable?
 - How sensitive are results to perturbation of model?
4. If necessary, change the model

Project ideas

See Google Sheets Schedule previous year proposals too!

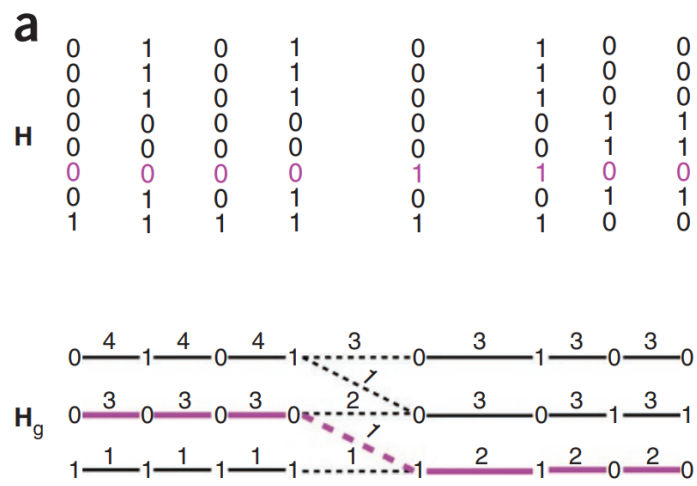
Data and models

- Dyadic data
 - Measured on pairs, e.g. social networks, user-movie rating data
 - Example model: matrix factorization
- Grouped data
 - Data belonging to a set of groups, e.g. documents (groups of words), DNA sequences (groups of populations)
 - Example models: Hierarchical models, mixed-membership
- Sequential and spatial data
 - Data measured over time, in a sequence, or over space, e.g. medical signals, DNA sequences, density of tree cover in a forest
 - Example models: HMMs, Kalman filter

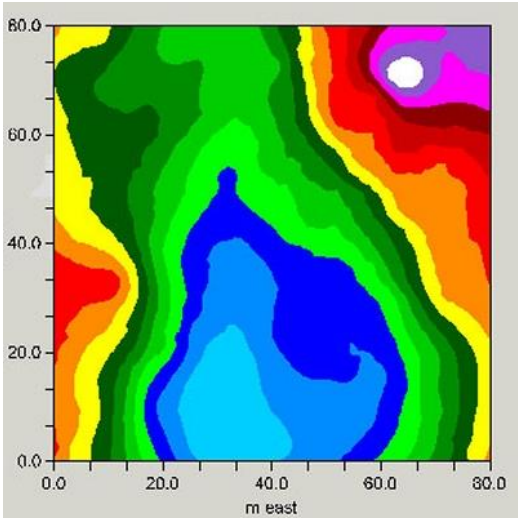
				
John 	5	1	3	5
Tom 	?	?	?	2
Alice 	4	?	3	?

Data and models

sequential
data



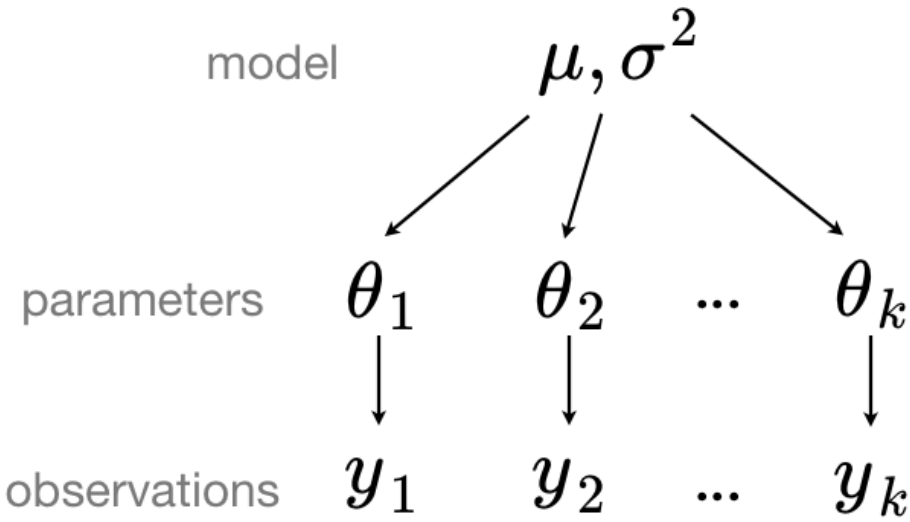
spatial data



Dyadic data

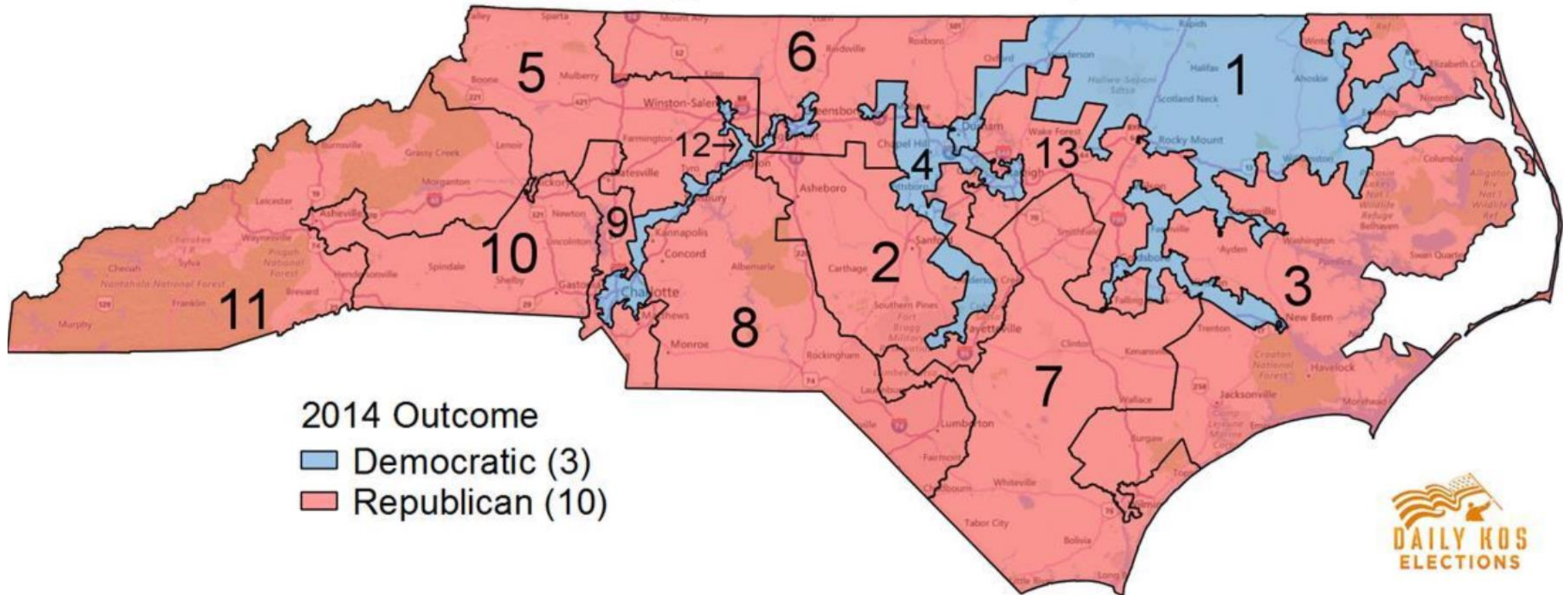


Grouped data



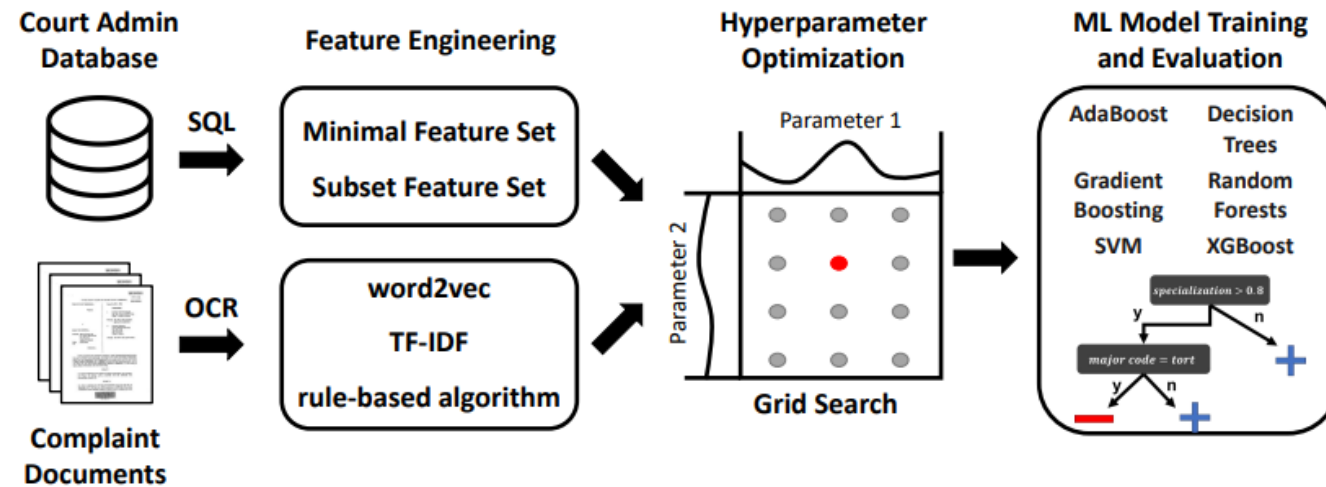
Project examples (politics)

- Modeling of U.S. house voting districts



Project examples (law)

- Modelling legal documents.
 - Given an active case file, find similar decided cases (help build a case)
 - Classify decisions as biased or not (related to above)
 - Predicting the growth of class of law professions (better allocation of resources)
 - Predict legal outcomes (should I settle or not?)



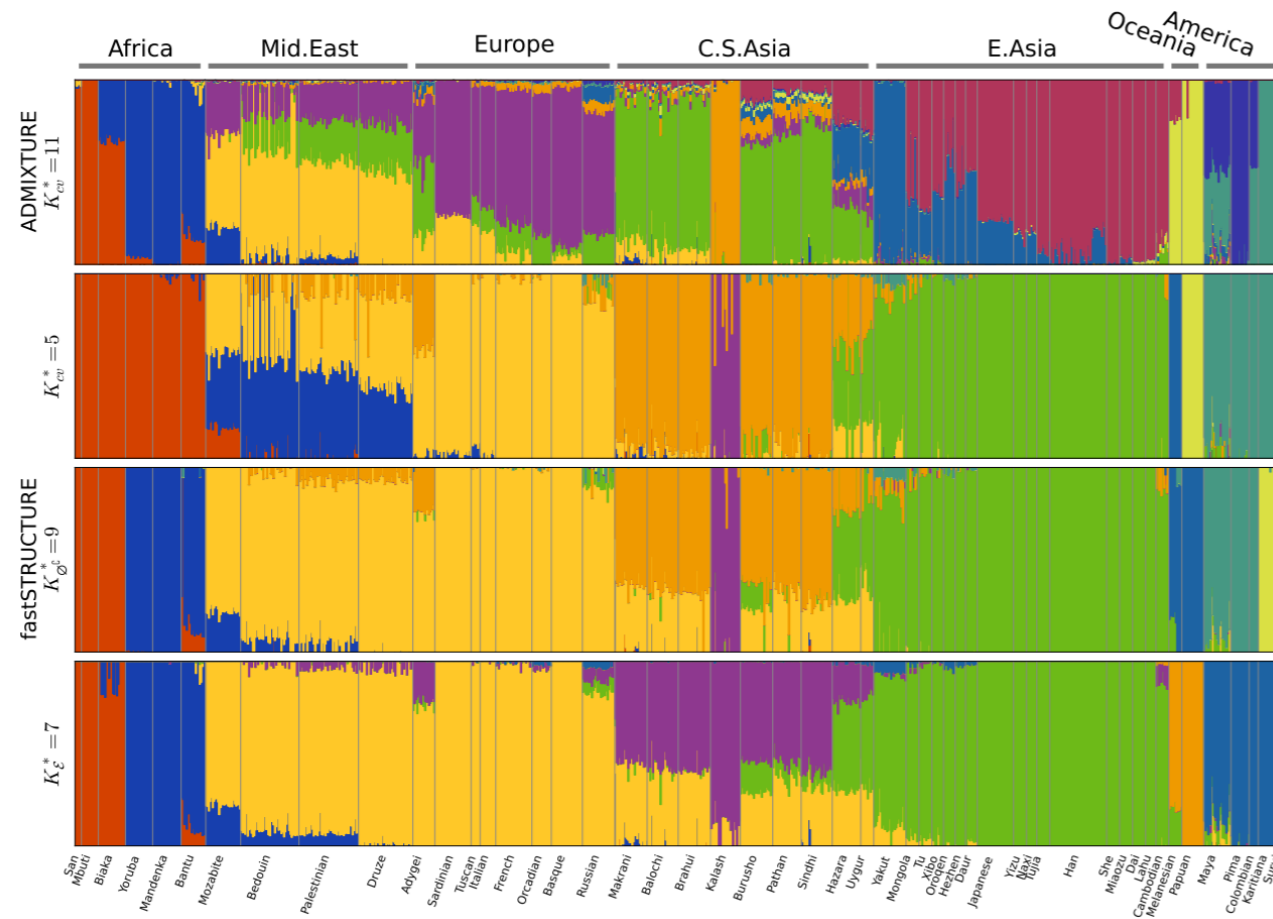
Project examples (genomics)

- Imputing of missing variation in genotype data

Reference				Observation	Prediction	
A	A	A	G	A/G	A	G
A	T	A	A	A/A	A	A
T	T	G	T	./.	T	T
G	G	G	G	./.	G	G
A	G	A	A	A/A	A	A
T	T	T	T	T/T	T	T
C	G	G	C	C/G	C	G

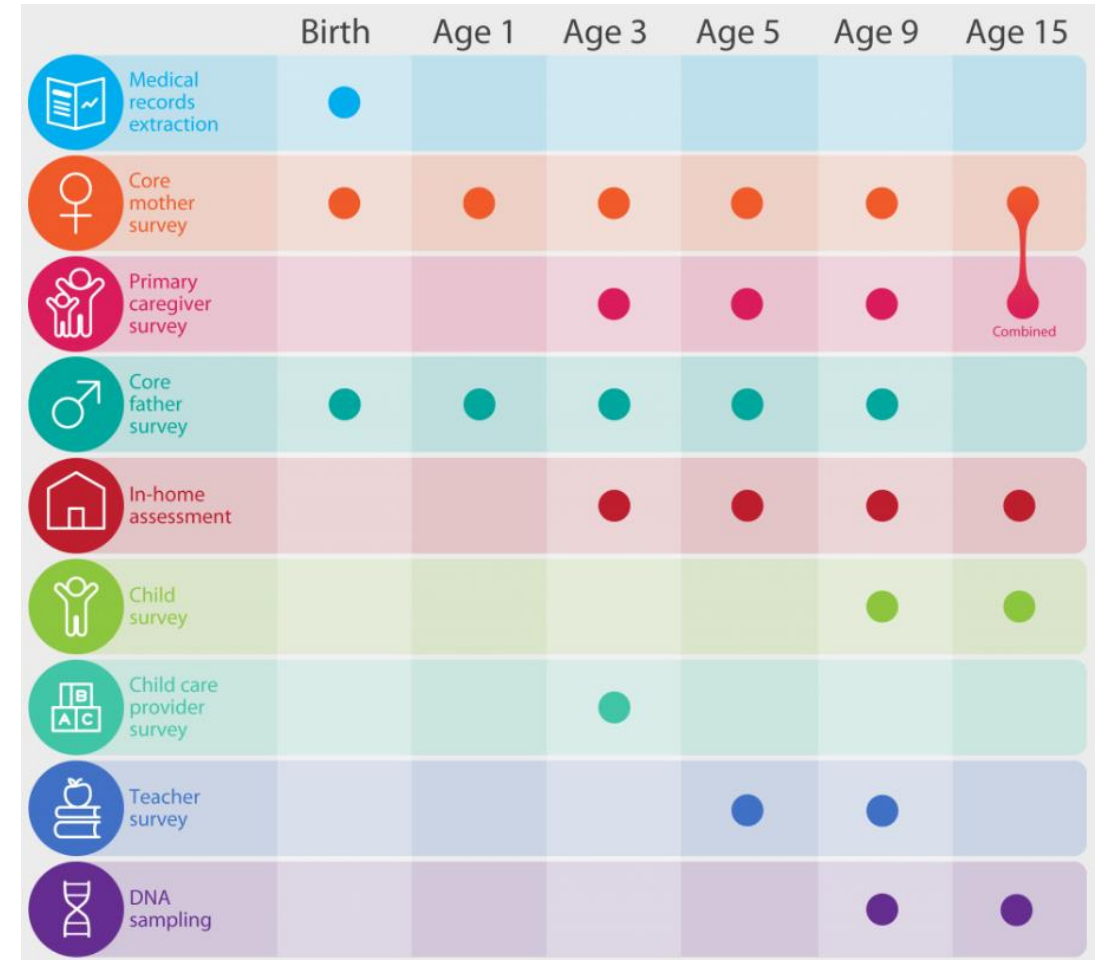
Project examples (evolutionary biology)

- Inferring local ancestry in genotype data (clustering of unknown populations).



Fragile Families data (sociology)

- ~5k families at socioeconomic risk
 - High rates of divorce, eviction, lack of access to economic opportunities
- Types of problems
 - Predict outcomes at age 15
 - Finding meta-predictors (feature selection)
 - Non-missing at random imputation



See course schedule for more

- Our work will build on **soccer match data** (passes, possession time, shots on goal, penalties, etc.) to develop a model that **predicts the final score of a soccer match**. We will evaluate several hypotheses, e.g. do home teams have an advantage? is there a 'buff' for attacking or defending teams at home?
- We will use the **Sloan Digital Sky Survey data** to **classify objects based on light spectrum**? Can we classify objects based on image features?
- We are interested in **estimating haplotypes** (sequences of co-inherited genetic variant alleles) from **population based genetics data** using Bayesian nonparametric and HMM models.
- We are interested in the **prediction of AirBnB prices** in New York City using **data from listings, including room types, reviews, and geographical information**.
- We are interested in modelling **league of legends data** from Riot's API. We will build models for **predicting winning teams and estimating player ELO rating**.